



MagDiff™ MA900B

Fully Differential, Magnetic
Digital Angle Sensor with Stray-Field Immunity

PRELIMINARY SPECIFICATION SUBJECT TO CHANGE

DESCRIPTION

The MA900B is a contactless, precision magnetic absolute angle position sensor. The angle is extracted from the magnetic field difference at several sensing locations on the IC. This differential method eliminates the effect of parasitic magnetic fields and is suitable for sensors positioned at the end-of-shaft with a simple dipole target magnet. Fast data acquisition and processing provide accurate angle measurement up to very high speeds (e.g. lag < 1° at 100'000 rpm).

The MA900B features magnetic field strength detection with programmable thresholds to allow sensing of the magnet position relative to the sensor for creation of functions such as sensing of axial movements, or for diagnostics.

On-chip, non-volatile memory provides storage for configuration parameters, including the reference zero angle position, ABZ encoder settings, configuration of UVW pole pair emulation and magnetic field detection thresholds.

FEATURES

- Absolute Angle Encoder
- $\pm 0.01^\circ$ RMS Noise at 1.5kHz Bandwidth
- Robust Against Parasitic Stray Fields
- Latency < 1 μ s
- Digital Angle Readout and Chip Configuration via SPI or I²C
- Digital Angle Readout via SENT and SSI
- ABZ Quadrature Encoder Interface with Programmable Pulses-Per-Turn (PPT)
- UVW Interface with 1-to-8 Pole Pair Emulation
- Programmable Magnetic Field Strength Detection for Diagnostic Checks
- Push-Button Interface
- 3.3V & 5V Supply, 11 mA Operating Current
- -40°C to +150°C Operating Temperature
- Available in a QFN-16 (3mmx3mm) Package with Wettable Flanks

APPLICATIONS

- Motor Control
- Human-Machine Interface (HMI)
- Robotics
- Industrial Control & Automation

All MPS parts are lead-free, halogen-free, and adhere to the RoHS directive. For MPS green status, please visit the MPS website under Quality Assurance. "MPS", the MPS logo, and "Simple, Easy Solutions" are trademarks of Monolithic Power Systems, Inc. or its subsidiaries.

TYPICAL APPLICATION

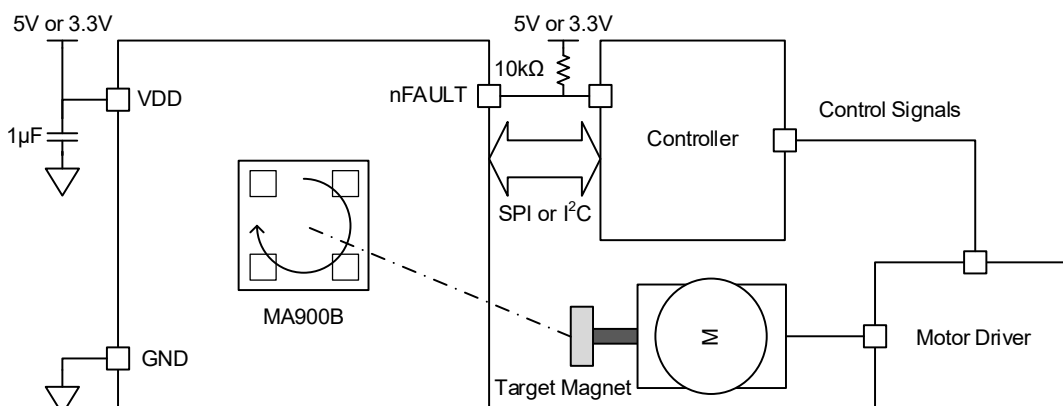


Figure 1: Typical Application


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ORDERING INFORMATION

Part Number*	Package	Top Marking	MSL Rating	Version
MA900BGQE-S-0000	QFN-16 (3mm x 3mm)	See Below	1	SPI - Default
MA900BGQE-C-0000	QFN-16 (3mm x 3mm)	See Below	1	I ² C - Default
MA900BGQE-S-xxxx**	QFN-16 (3mm x 3mm)	See Below	1	SPI - Customized Registers (contact factory)
MA900BGQE-C-xxxx**	QFN-16 (3mm x 3mm)	See Below	1	I ² C - Customized Registers (contact factory)
EVKT-MagAlpha-MagDiff	Evaluation kit	N/A	N/A	N/A

* For Tape & Reel, add suffix -Z (e.g., MA900BGQE-C-xxxx-Z).

** “xxxx” is the configuration code identifier for the register settings. The first four digits of the suffix (“xxxx”) can be a hexadecimal value between 0 and F. Work with an MPS FAE to create this unique number for any non-default function option. “0000” is the factory default code.

TOP MARKING

TBD

PACKAGE REFERENCE

TOP VIEW SPI Version (MA900BGQE-S-xxxx) QFN-16 (3mm x 3mm)	TOP VIEW I²C Version (MA900BGQE-C-xxxx) QFN-16 (3mm x 3mm)
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PRELIMINARY SPECIFICATION SUBJECT TO CHANGE**PIN FUNCTIONS**

Name	Package Pin		Description
	SPI Version	I ² C Version	
VDD	1	1	Voltage supply. Connect to 5V or 3.3V. A 1μF, low-ESR capacitor is required between VDD and GND. See the Electrical Mounting and Power Supply Decoupling section on page 56 for additional information.
NC	2	2	No connection. Leave this pin floating.
GND	3	3	Ground.
CIPO	4		SPI controller input peripheral output (CIPO). Output. The pin is push-pull when the SPI is active. When the SPI is idle, the pin configuration is determined by the SPIOPULL bit, see the SPI Pins Configuration section on page 20.
SA0		4	I²C serial address definition. At start-up, the SA0, SA1 and SA2 pins voltage defines the I ² C Sensor Address, see the I ² C Sensor Address section on page 22.
COPI	5		SPI controller output peripheral input (COPI). Input. Pin configuration is determined by the SPIIPULL bit, see the SPI Pins Configuration section on page 20. Connect to GND if not used.
SDA		5	I²C serial data. Input-Output, open-drain. Connection to VDD via a pull-up resistor is required. Recommended value is 1.5 kΩ.
SCLK	6		SPI serial clock. Input. Pin configuration (Hi-Z or pull-down) is determined by the SPIIPULL bit, see the SPI Pins Configuration section on page 20. Connect to GND if not used.
SCL		6	I²C serial clock. High-impedance input. Connection to VDD via a pull-up resistor is required. Recommended value is 1.5 kΩ.
nCS	7		SPI chip select. Active-low input. Pin configuration is determined by the SPIIPULL bit, see the SPI Pins Configuration section on page 20. Pull up to VDD via a 10k resistor if not used.
SA1		7	I²C serial address definition. At start-up, the SA0, SA1 and SA2 pins voltage defines the I ² C Sensor Address, see the I ² C Sensor Address section on page 22.
IO0	8	8	Multi-function I/O. Pin function is determined by the IOM bits, see the IO Pins Function Selection section on page 50. Leave floating if not used.
IO1	9	9	Multi-function I/O. Pin function is determined by the IOM bits, see the IO Pins Function Selection section on page 50. Leave floating if not used.
IO2	10	10	Multi-function I/O. Pin function is determined by the IOM bits, see the IO Pins Function Selection section on page 50. Leave floating if not used.
nFAULT	11	11	Fault output. Active-low output, open-drain. Connect to VDD via 10kΩ pull-up resistor.
IO3	12	12	Multi-function I/O. Pin function is determined by the IOM bits, see the IO Pins Function Selection section on page 50. Leave floating if not used.
IO4	13	13	Multi-function I/O. Pin function is determined by the IOM bits, see the IO Pins Function Selection section on page 50. Leave floating if not used.
IO5	14	14	Multi-function I/O. Pin function is determined by the IOM bits, see the IO Pins Function Selection section on page 50. Leave floating if not used.
IO6	15	15	Multi-function I/O. Pin function is determined by the IOM bits, see the IO Pins Function Selection section on page 50. Leave floating if not used.
NU	16		Not used. Connect to GND.
SA2		16	I²C serial address definition. At start-up, the SA0, SA1 and SA2 pins voltage defines the I ² C Sensor Address, see the I ² C Sensor Address section on page 22.



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PRELIMINARY SPECIFICATION SUBJECT TO CHANGE**ABSOLUTE MAXIMUM RATINGS** ⁽¹⁾

VDD pin	-0.5V to +7.0V
NC pin	-0.3V to +3.3V
All other pins	-0.5V to +7.0V
Junction temperature (T _J)	165°C
Lead temperature	260°C
Storage temperature	-65°C to 150°C

ESD Ratings

Human body model (HBM) ⁽²⁾	±2kV
Charged-device model (CDM) ⁽³⁾	±750V

Recommended Operating Conditions ⁽⁴⁾

V _{DD} , 3.3V operation	3V to 3.6V
V _{DD} , 5V operation	4.5V to 5.5V
Non-supply pins voltage	-0.3V to V _{DD} +0.3V
Operating junction temp.	-40°C to +150°C
Operating magnetic field	10mT to 200mT
Homogeneous stray field	up to 1000mT
MTP Store operation temp.	-40°C to +85°C

Thermal Resistance ⁽⁵⁾ **R_{JA}** **R_{JB}**

JESD51-7, 4-layer PCB	58	12	°C/W
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Notes:

- 1) Exceeding these ratings may damage the device.
- 2) Measured according to JEDEC JS-001
- 3) Measured according to JEDEC JS-002
- 4) The device is not guaranteed to function outside of its operating conditions.
- 5) $R_{JA} = (T_J - T_A)/P_D$ and $R_{JB} = (T_J - T_B)/P_D$. T_A, T_B and P_D are, respectively, the ambient temperature, the PCB temperature in proximity of the device and the power dissipation.



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PRELIMINARY SPECIFICATION SUBJECT TO CHANGE**ELECTRICAL CHARACTERISTICS** *$V_{DD} = 3.3V$ or $5V$, $T_J = -40^{\circ}C$ to $+150^{\circ}C$, typical values are at $25^{\circ}C$, unless otherwise noted.*

Parameter	Symbol	Condition	Min	Typ	Max	Units
Power Supply						
Supply current	I _{DD}	T _J = 25°C		11	12.5	mA
		T _J = -40°C to +150°C			15	mA
Digital I/O (3.3V operation)						
Input						
Input high voltage	V _{IH}		V _{DD} ×0.7			V
Input low voltage	V _{IL}				V _{DD} ×0.3	V
Input leakage current	I _{LK}	Except Pin 16			100	nA
		Pin 16			100	μA
Input series resistor	R _{ESD}			1.5		kΩ
Pull-up resistor	R _{PU}		15		115	kΩ
Pull-down resistor	R _{PD}		15		160	kΩ
Output - Push-pull						
Output low voltage	V _{OL}	DRV = 0, I _{OUT} = 4mA		0.3	0.55	V
Output high voltage	V _{OH}	DRV = 0, I _{OUT} = 4mA	2.5	2.8		V
Rising edge slew rate	T _R	DRV = 0, C _L = 50pF		0.1		V/ns
Falling edge slew rate	T _F	DRV = 0, C _L = 50pF		0.15		V/ns
Output - Open-drain						
Output low voltage	V _{OL}	DRV = 0, I _{OUT} = 4mA		0.3	0.55	V
Open drain leakage current	I _{LK}				100	nA
Digital I/O (5V operation)						
Input						
Input high voltage	V _{IH}		V _{DD} ×0.7			V
Input low voltage	V _{IL}				V _{DD} ×0.3	V
Input leakage current	I _{LK}	Except Pin 16			100	nA
		Pin 16			150	μA
Input series resistor	R _{ESD}			1.5		kΩ
Pull-up resistor	R _{PU}		8		80	kΩ
Pull-down resistor	R _{PD}		8		100	kΩ
Output - Push-pull						
Output low voltage	V _{OL}	DRV = 0, I _{OUT} = 4mA		0.2	0.4	V
Output high voltage	V _{OH}	DRV = 0, I _{OUT} = 4mA	4.5	4.7		V
Rising edge slew rate	T _R	DRV = 0, C _L = 50pF		0.2		V/ns
Falling edge slew rate	T _F	DRV = 0, C _L = 50pF		0.3		V/ns
Output - Open-drain						
Output low voltage	V _{OL}	DRV = 0, I _{OUT} = 4mA		0.2	0.4	V
Open drain leakage current	I _{LK}				100	nA



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PRELIMINARY SPECIFICATION SUBJECT TO CHANGE**GENERAL CHARACTERISTICS**

$V_{DD} = 3.3V$ or $5V$, $30 \text{ mT} \leq B_z \leq 100 \text{ mT}$, $T_J = -40^\circ\text{C}$ to $+150^\circ\text{C}$, typical values are at 25°C , unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Absolute Output – Serial Outputs						
Data output length					16	bit
Resolution ⁽⁷⁾		B _z = 48mT, FW = 8, T _J = 25°C	11.5	12.4		bit
		B _z = 48mT, FW = 8, T _J = -40°C to +150°C		12.2		bit
RMS noise ⁽⁷⁾		B _z = 48mT, FW = 8, T _J = 25°C		0.011	0.018	deg
		B _z = 48mT, FW = 8, T _J = -40°C to +150°C		0.013		deg
Refresh rate	f _{REFRESH}	T _J = 25°C	970	1000	1030	kHz
Response time						
Power-up time		FW = 8			2.5	ms
Front-end time constant	T _{FE}			0.75		μs
Digital propagation delay	T _{DP}			6		μs
Total latency ⁽⁶⁾	L	FW = 0		9		μs
		FW > 0		0	1	μs
Filter cutoff frequency	f _{CUTOFF}	FW = 8		1560		Hz
Accuracy						
Integral nonlinearity ⁽⁷⁾	INL	B _z = 48mT, T _J = 25°C		0.38	0.7	deg
		B _z = 48mT, T _J = -40°C to +150°C		0.38	0.74	deg
Temperature induced variation ⁽⁷⁾		B _z = 48mT, T _J = -40°C to +125°C	-0.62	±0.29	0.62	deg
		B _z = 48mT, T _J = -40°C to +150°C	-0.7	±0.32	0.7	deg
Voltage supply induced drift ⁽⁷⁾		B _z = 48mT, T _J = 25°C, VDD = 3.3V	-0.9	±0.2	0.9	deg/V
		B _z = 48mT, T _J = 25°C, VDD = 5V	-0.13	±0.04	0.13	deg/V
Homogeneous stray field influence ⁽⁷⁾		B _z = 48mT, T _J = 25°C			0.034	°INL/mT
Zero Setting						
Default zero tolerance	δ _{ZERO}			±7.5		deg
Absolute Output – SENT						
SENT angle data length		Configurable	12		16	bit
SENT tick length		Configurable	0.25		24	μs
SENT falling edge slew rate		Configurable	1		128	V/μs
Absolute Output – PWM						
PWM Frequency	f _{PWM}	PWMF = 0		970		Hz
		PWMF = 1		242.5		Hz
Incremental Output – ABZ						
ABZ update rate				16		MHz
ABZ edges per turn		Configurable	4		8192	
ABZ Pulses per channel per turn	PPT+1	Configurable	1		2048	



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PRELIMINARY SPECIFICATION SUBJECT TO CHANGE**GENERAL CHARACTERISTICS (continued)**

$V_{DD} = 3.3V$ or $5V$, $30\text{ mT} \leq B_Z \leq 100\text{ mT}$, $T_J = -40^\circ\text{C}$ to $+150^\circ\text{C}$, typical values are at 25°C , unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
ABZ hysteresis	H	Configurable	0.0055		15.9	deg
ABZ jitter (peak-peak)		PPT = 511, speed = 1krpm		± 0.046		deg
Incremental Output – UVW						
UVW cycles per turn	NPP+1	Configurable	1		8	
UVW hysteresis	H	Configurable	0.0055		15.9	deg
UVW jitter (peak-peak)		NPP = 0, speed = 1krpm		± 0.23		deg
Speed Output						
Speed scale	S_{SPEED}			3.576		rpm/LSB
Speed output error ⁽⁷⁾		FWS = 7, speed = 10krpm, $T_J = 25^\circ\text{C}$			3	%
Speed RMS noise		FWS = 7		125		rpm
Temperature output						
Temperature output sensitivity	S_T			8		LSB/ $^\circ\text{C}$
Magnetic Field Detection Thresholds						
Offset				1		mT
Gain error				3		%
Temperature drift				± 300		ppm/ $^\circ\text{C}$
MTP Memory Operations						
MTP Store time	t_{STORE}			650		ms
MTP Restore time	$t_{RESTORE}$			55		μs

Notes:

6) Guaranteed by design.

7) Guaranteed by characterization. The min and max values are defined as the mean value $\pm 3\sigma$.

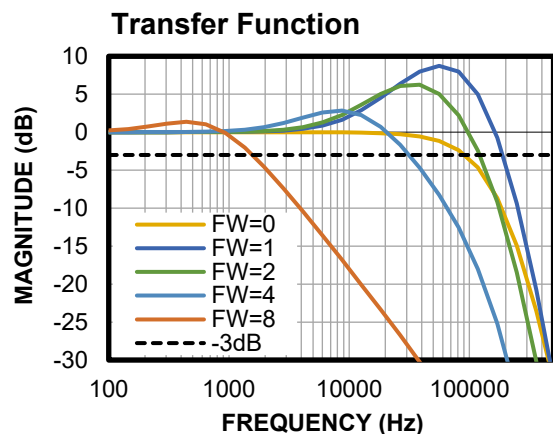


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TYPICAL CHARACTERISTICS

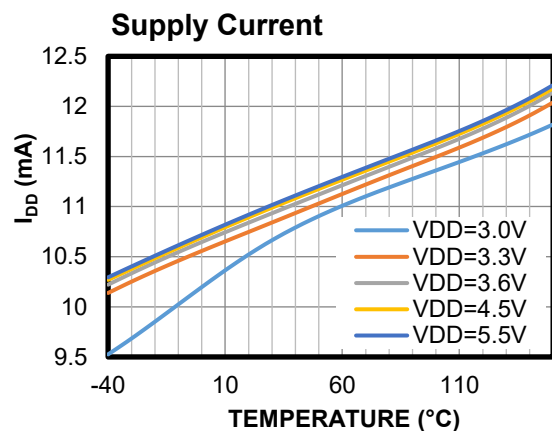
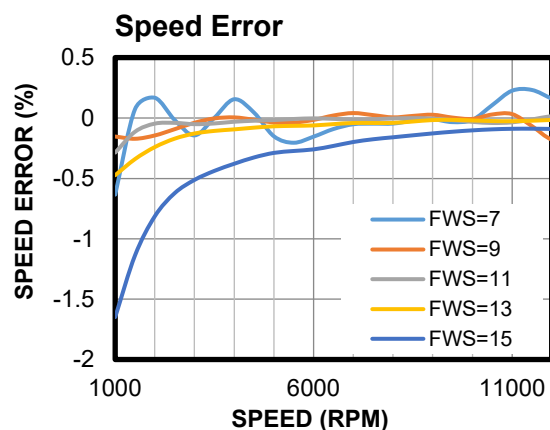
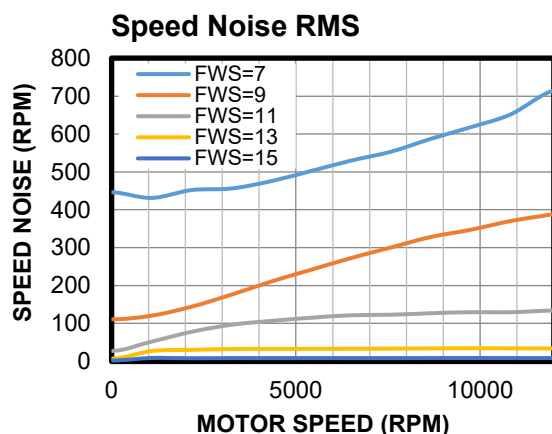
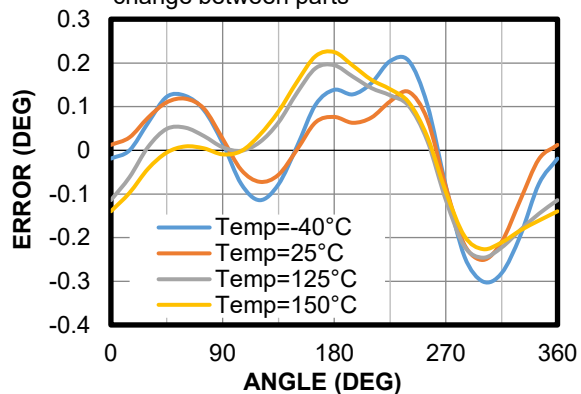
$V_{DD} = 3.3V$ or $5V$, $30\text{ mT} \leq B_z \leq 100\text{ mT}$, $T_J = 25^\circ\text{C}$, unless otherwise noted.



Error Curves

$V_{DD} = 5V$, $B = 48\text{mT}$

Phase and Amplitude
change between parts





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BLOCK DIAGRAM

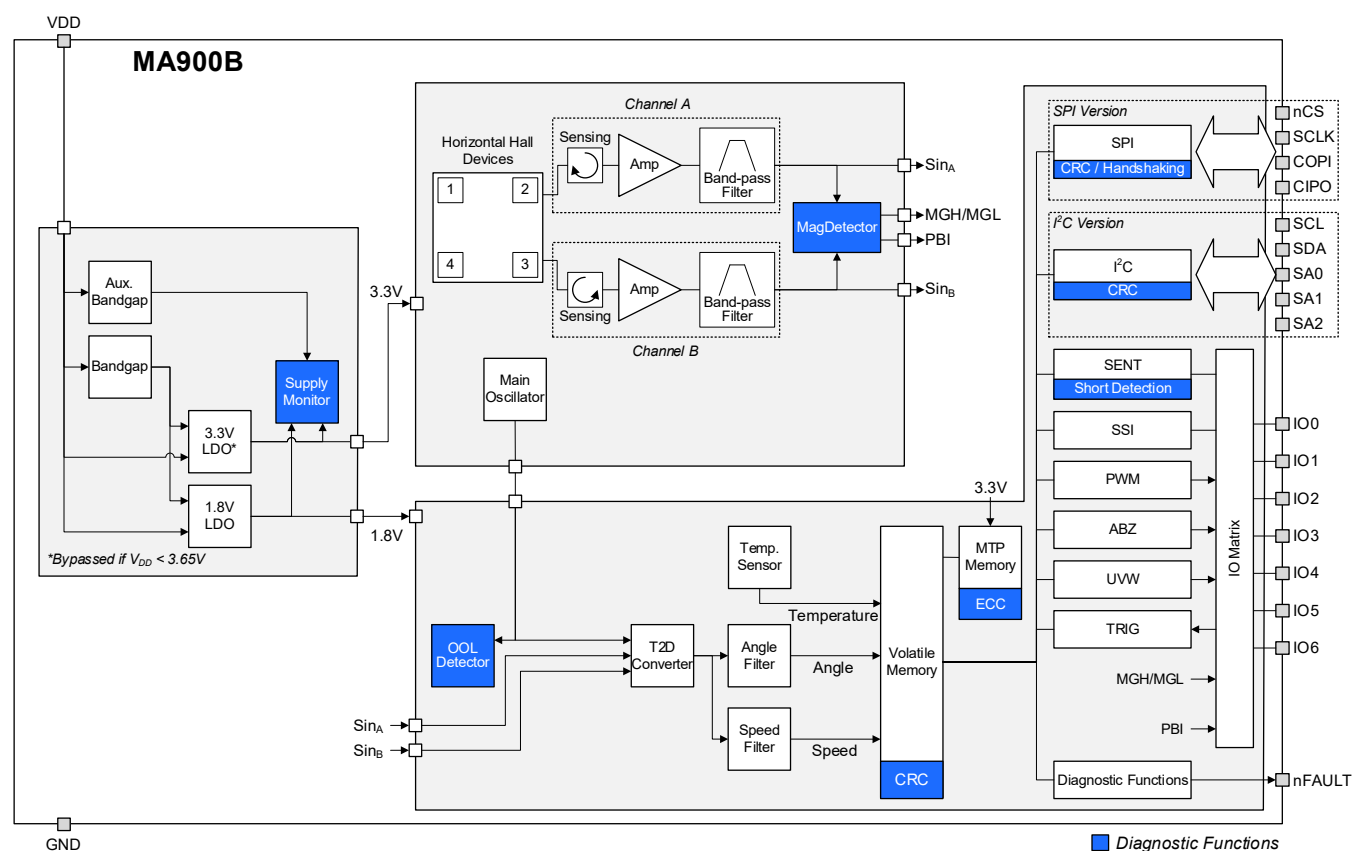


Figure 2: Block Diagram



OPERATION

Sensor Front-End

The magnetic field in the sensor Z direction (Figure 3) is detected by four integrated Horizontal Hall devices located in the corners of the die, on a 2.2 mm diameter circle (Figure 4, Figure 5).

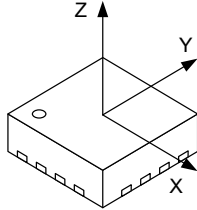


Figure 3: Sensor Coordinate System

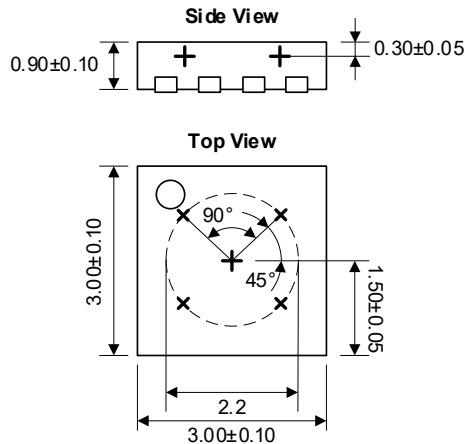


Figure 4: Hall Devices Position

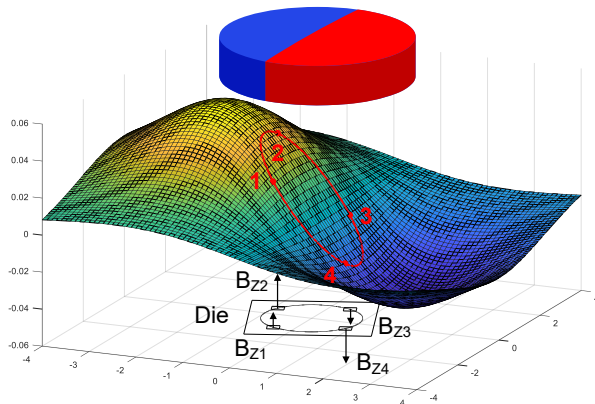
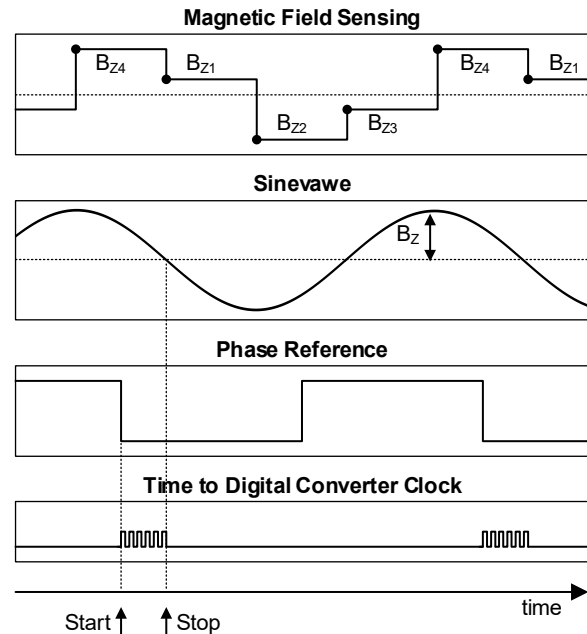


Figure 5: Magnetic Field in the Z Direction (3D surface) generated by a Cylinder Magnet and Field Detected by the Hall Devices (B_{z1} to B_{z4})

The angle is measured using the Spinaxis™ method, which digitizes the direction of the field without complex arctangent computation or feedback loop-based circuits (interpolators).

The four Hall devices are probed sequentially (Figure 6 – Magnetic Field Sensing) and the resulting signal is filtered to get a smooth sinewave (Figure 6 – Sinewave). The phase of the sinewave is proportional to the angle of the magnet and is not affected by homogeneous stray fields. The phase is digitized through a time-to-digital (T2D) converter (see Figure 6 – Time to Digital Converter Clock).



**Top: Magnetic Field Samples
Mid: Fundamental Harmonic
Bottom: Clock of Time-to-Digital Converter
Figure 6: Phase Detection Method**

This process is run in parallel in two measurement channels, A and B, see Figure 2 on page 10. Channel A and channel B differ in the probing sequence of the Hall devices: 1, 2, 3, 4 for channel A and 1, 4, 3, 2 for channel B.

The two channels generate the $\sin_A$ and $\sin_B$ sinewaves, whose phases, ϕ_A and ϕ_B , are ideally opposites ($\phi_A = -\phi_B$). The digitized angle, α , is calculated as:

$$\alpha = \frac{\phi_A - \phi_B}{2} \quad (1)$$

Note: the quantity B_z shown in Figure 6 is essentially the gradient of the magnetic field at the sensor center multiplied by the radius of the circle where the Hall devices are located



(1.1 mm). In this document, we refer to B_z , the field amplitude in T, instead of the actual gradient in T/m, because B_z is the quantity actually measured by the sensor and thus the one determining the sensor performances.

Sensor – Magnet Mounting

The differential measurement technique used in MA900B only works in an end-of-shaft configuration, namely if the sensor center is aligned with the magnet rotation axis (see Figure 7).

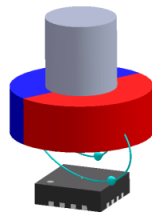


Figure 7: End-of-Shaft Mounting

Homogeneous Stray-Field Immunity

The measurement principle used in MA900B provides immunity to homogeneous stray fields.

A homogeneous stray field causes the signals of the four Hall devices to shift by the same offset. This offset does not modify the sinewave (see Figure 6), from which angle measurement is extracted.

Digital Conditioning

The digitized angle value is processed by a unit containing the digital filtering, zero and rotation direction setting.

After the digital conditioning, the angle value is stored in the ANGLE0 and ANGLE1 registers, where it can be read via SPI, I²C, SENT or SSI interface. This value is also used to generate the ABZ, UVW, PWM outputs.

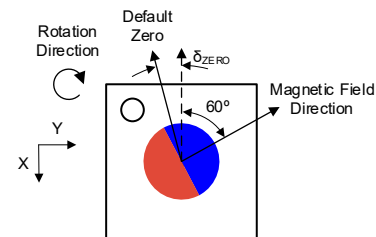
Digital Filtering

Digital filtering is used to reduce the measurement noise and compensate for the latency introduced by the measurement process. The digital filter is configurable: this allows to adjust the measurement noise and bandwidth to better match the requirements of each application. See the Position Output Filter section on page 49 for additional information about the filter configuration.

Zero and Rotation Direction

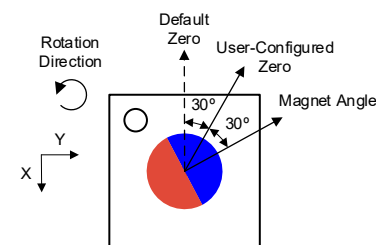
The output angle is referenced to the sensor zero and its sign is determined by the sensor rotation direction. Both these parameters are user configurable, see the ZERO1 register description on page 33 and the DIR register description on page 39 for additional details.

By default, the sensor zero is aligned with the package X axis, with a tolerance δ_{ZERO} (see the General Characteristics table on page 7), and, when looking at the sensor top surface, the rotation direction is clockwise. Figure 8 and Figure 9 show respectively the output angle with default and custom programming for the same magnet angle.



$$\text{Output Angle} = 60^\circ + \delta_{ZERO}$$

Figure 8: Sensor Output with Default Zero and Default Rotation Direction



$$\text{Output Angle} = -30^\circ = 330^\circ$$

Figure 9: Sensor Output with user configured Zero = 30° and Counterclockwise Rotation Direction ($\delta_{ZERO} = 0$)

Speed Measurement

The digitized angle value is used to calculate the angular speed of the magnet. Similarly to the angle, the speed value is processed by a configurable digital filter. The filtered speed value is stored in the SPEED0 and SPEED1 registers and it can be read via SPI, I²C or SENT interface.

Multiturn Measurement

Whenever the angle measurement is crossing 0°, an internal multiturn counter is incremented or



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decremented according to the direction of the crossing. The counter value is stored in the MTV0 and MTV1 registers and it can be read via SPI, I²C or SENT interface.

The counter value can be offset by writing in the MTV0 and MTV1 registers. See the MTV1 register description on page 42 for additional details.

Temperature Measurement

The MA900B is equipped with a temperature sensor. The temperature measurement is stored

in the TEMP0 and TEMP1 registers and it can be read via SPI, I²C or SENT interface.

Digital Interfaces

MA900B provides multiple digital interfaces, which are available on the IO0 to IO6 pins. They are summarized in Table 1.

MA900B supports both 3.3V and 5V operation. The levels of the digital input/output pins are determined by the supply voltage.

Table 1: Digital Interfaces Supported on MA900B

Interface	Description	See Page
ABZ	Emulates the output of an incremental optical encoder	50
UVW	Emulates the output of three Hall switches	51
PWM	Provides a fixed frequency square wave signal whose duty cycle changes with the angle	52
MGH/MGL (MagHigh/MagLow)	Signals when the magnetic field is above or below a programmable threshold	53
PBI (Push Button Interface)	Signals an increase/decrease of the magnetic field	54
TRIG (Trigger)	Input to latch the Angle, Speed, Multiturn, Temperature measurements	55
GPIO (General Purpose I/O)	IO0 to IO6 pins can be used as digital inputs or outputs	55

Electrical Test Mode

The Electrical Test (ET) mode allows verifying the integrity of the signal path. When enabled, the signal from the Hall devices output is replaced by a configurable reference signal which corresponds to a specific angle output. The correct operation of the signal path can be verified by comparing the expected and measured output.

The ET mode is enabled and configured via the ETEST register. See the ETEST register description for more details.

Diagnostic Functions

The MA900B supports multiple Diagnostic Functions to detect fault conditions during its operation.

Whenever a fault condition is triggered, the following operations are performed:

- The nFAULT pin is pulled to GND.
- The flag corresponding to the fault condition is raised in the AFAIL, DFAIL or IFAIL register.

- If the raised flag is in the AFAIL register, the AFAIL flag in the STATUS register is also raised. The same applies to DFAIL and IFAIL registers/flags.
- For SPI version, if the SPI CRC inversion is enabled (CRCINVEN=1), the SPI CRC4 sent on CPO is inverted.

All fault related flags are sticky, namely they are not automatically cleared if the fault condition is removed. The flags must be cleared by writing a 1 in the corresponding bit. The flag is not cleared if the fault is still persistent.

Supply Voltage Monitor

The internal 3.3V and 1.8V supply lines are continuously monitored to detect undervoltage and overvoltage conditions. If an undervoltage or overvoltage is detected on the 3.3V (1.8V) line, a fault condition is triggered and the ASUPPLY (DSUPPLY) flag is asserted in the AFAIL register.

**Phase-Locked Loop (PLL) Out Of Lock (OOL) Detector**

The main clock is generated by a PLL circuit. The PLL residual phase error is continuously monitored. If the phase error is outside a valid range, a fault condition is triggered and the OOL flag in the AFail register is raised.

MagDetector

The amplitude of the Sin_A and Sin_B signals is proportional to the field amplitude B_z and it is continuously monitored to ensure that B_z is in the expected range. See the MGH/MGL section on page 53 for more information.

Register CRC

A Cyclic Redundancy Check (CRC) algorithm is used to detect errors in the volatile memory. If an

error is detected, a fault condition is triggered and the REGCRC flag in the DFail register is raised.

MTP Memory ECC

An Error Correction Code (ECC) algorithm is used to detect error in the MTP memory. See the MTP Flags section on page 25 for more information.

Communication Error Detection

The SPI, I²C and SENT interfaces integrate safety mechanisms to detect communication errors. They are described in the SPI, I²C and SENT sections.



SERIAL INTERFACES

MA900B supports 4 serial communication interfaces: SPI, I²C, SSI and SENT. SPI and I²C can be used both to read data from the sensor and for sensor configuration; while SENT and SSI can only be used to read data.

SPI

SPI is a four-wire, synchronous, serial communication interface. MA900B supports SPI Mode 0 and Mode 3 (see Table 2 and Table 3). The SPI Mode is detected automatically by the sensor and therefore does not require any action from the user. The maximum SPI clock frequency supported by MA900B is 20MHz, and there is no minimum clock rate. See Figure 10, Figure 11 and Table 4 for details on SPI timing.

Note that actual data rates depend on the PCB layout quality and signal trace length. The SPI Signals Routing on PCB section on page 56 provides some PCB layout recommendations.

In this document, COPI stands for Controller Output Peripheral Input, CIPO stands for Controller Input Peripheral Output.

All operations to the MA900B, whether for writing or reading a register content, must be transferred through the SPI COPI pin.

Table 2: SPI Specification

	Mode 0	Mode 3
SCLK Idle State	Low	High
Data Capture	On SCLK rising edge	
Data Transmission	On SCLK falling edge	
nCS Idle State	High	
Data Order	MSB first	

Table 3: SPI Standard

	Mode 0	Mode 3
CPOL	0	1
CPHA	0	1
Data Order (DORD)	0 (MSB first)	

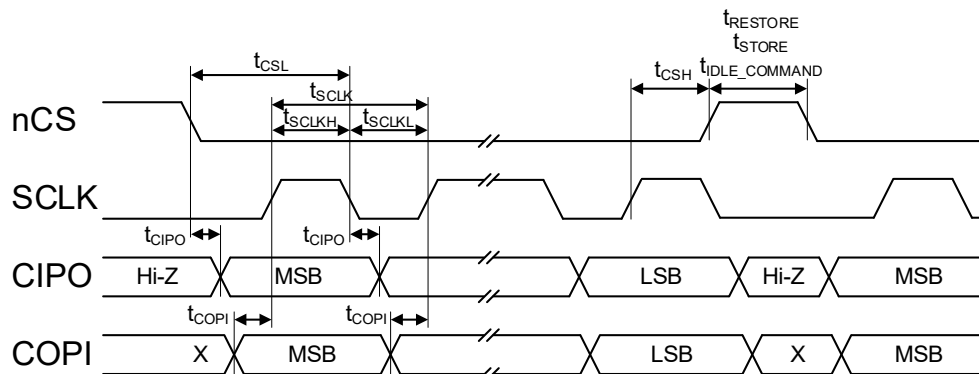


Figure 10: SPI Mode 0

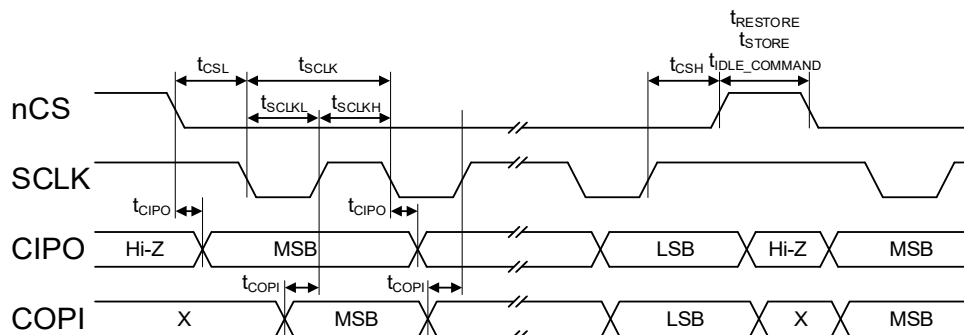


Figure 11: SPI Mode 3



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Table 4: SPI Timings

Parameter ⁽⁸⁾	Description	Min	Max	Unit
t _{IDLE_COMMAND}	Idle time between transmissions	100		ns
t _{CSL}	Time between nCS falling edge and SCLK falling edge	25		ns
t _{SCLK}	SCLK period	50		ns
t _{SCLKL}	Low level of SCLK signal	25		ns
t _{SCLKH}	High level of SCLK signal	25		ns
t _{CSH}	Time between SCLK rising edge and nCS rising edge	25		ns
t _{CIPO}	SCLK setting edge to data output valid		20	ns
t _{COPI}	Data input valid to SCLK reading edge	20		ns

Note:

8) All values are guaranteed by design

SPI Operations

MA900B supports two SPI operations:

- Read (one or multiple consecutive registers)
- Write (one or multiple consecutive registers)

Angle, Speed, Multiturn and Temperature measurements can be obtained by performing a Read operation on the corresponding registers.

MTP Store/Restore operations are executed by performing specific Write operations on the MTP register (see the MTP Memory Operations section on page 24).

SPI Read and Write operations always return the 16-bit angle at the beginning of operation, allowing to minimize the measurement latency.

There are two mechanisms for communication error detection:

- Data transmitted on COPI/CIPO can be protected with a 4-bit CRC code (CRC4) for every 12 bits of data. See APPENDIX A: CRC Calculation for CRC calculation details.
- A handshaking protocol can be activated for the transmitter to verify that the receiver got the correct data.

The MA900B can be configured to corrupt the SPI communication by inverting the CRC when a fault condition is detected.

SPI Operations Common Features

The SPI frames used for both Read and Write operations share some common characteristics:

- The communication is organized in 16-bit sized frames.
- The controller must send an operation code (OpCode) on the first 4 bits (MSBs) of the first 16-bit frame on COPI. This code is used to distinguish between a read and a write operation.
- In the following 8 bits of the first frame, the controller must send the address (Addr_N) of the register on which the read or write operation is performed.
- The last 4 bits of each 16-bit frame are reserved for the CRC4, which is calculated from the value of the 12 preceding bits. If the CRC is enabled, the controller must send a correct CRC4 value to avoid a communication error. The sensor always sends the CRC4 regardless of the value of CRCEN.
- The 16-bit angle value is sent by the sensor on CIPO in the first 12 bits of the first frame and the first 4 bits of the second frame, see Angle_{MSB} and Angle_{LSB} in Figure 12 on page 17. The 16-bit value is obtained as Angle_{MSB}*16 + Angle_{LSB}.
- If the SPI Handshaking is enabled (SPIHS = 1) and an SPI Read is being performed, starting from the second 16-bit frame the controller must send on COPI the 16-bits received on CIPO in the previous frame.



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- Regardless of SPI Handshaking status, when a SPI Write is being performed, the sensor sends on CIPO the 16-bits received on COPI in the previous frame (starting from the second 16-bit frame).

SPI Read

The SPI Read operation allows to read a single register or multiple consecutive registers. The

format for a single register read is shown in Figure 12:

- The OpCode to use for SPI Read operations is 0b0011
- In the second CIPO frame, Angle_{LSB} is followed by the 8-bit Data_N, namely the value of the register at the Addr_N address.

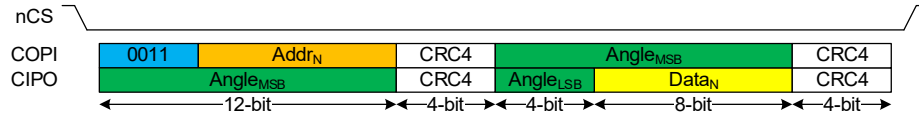


Figure 12: SPI Read Single

The SPI Read operation can be extended to read a sequence of consecutive registers (Read Burst).

In each additional frame i , the sensor sends the value of the register at the Addr_{N+i} address on

CIPO. The first 4 bits of each frame always contain Angle_{LSB}.

The format for a 2-registers Read Burst is shown in Figure 13.

The length of the SPI Read Burst is limited only by the number of available registers to read.

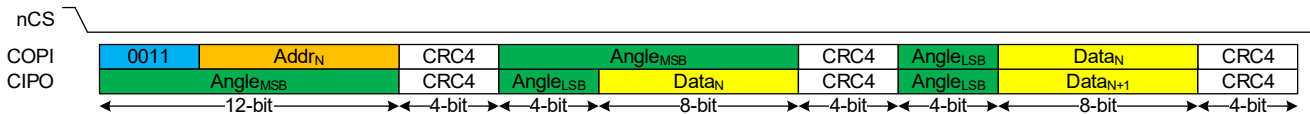


Figure 13: SPI Read Burst

SPI Write

The SPI Write operation allows to write a single register or multiple consecutive registers. The format for a single register write is shown in Figure 14:

- The OpCode to use for SPI Write operations is 0b1100.

- In the second COPI frame, the controller must send 0b0000, followed by the 8-bit data (Data_N) to write in the register.
- The 0b111100000000 code is transmitted via COPI to indicate the end of the SPI Write operation. All additional data transmitted on COPI after this code are ignored.

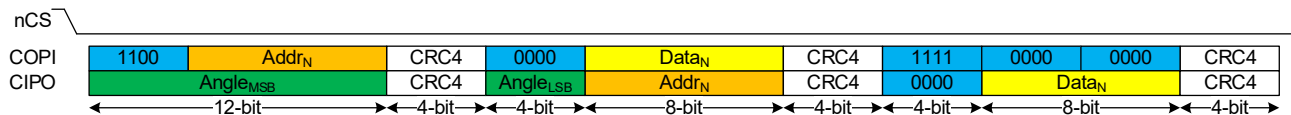


Figure 14: SPI Write Single

The SPI Write operation can be extended to write in a sequence of consecutive registers (Write Burst).

In each additional 16-bit frame i , the controller must send on COPI the value to write in the register at the Addr_{N+i} address.

As for single write, the controller must send via COPI the 0b111100000000 code after the last write value.

The SPI Write format for a 2-registers Write Burst is shown in Figure 15

The length of the SPI Write Burst is limited only by the number of available registers to write.

If the CRC is enabled (CRCEN = 1), the register value is changed only if the CRC4 sent by the controller is correct. If an invalid CRC4 is detected, the SPI protocol error flag is raised and the current write operation and all the following ones are ignored.



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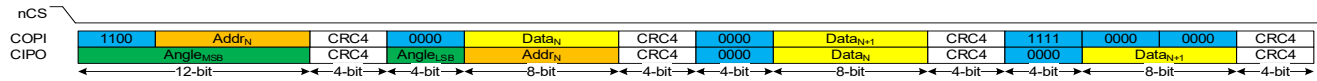


Figure 15: SPI Write Burst

SPI Read Angle

Any SPI Read or SPI Write operation can be used to read the 16-bit angle, as it is always sent by the sensor on COPI in the first two frames of each operation. Nonetheless, it is recommended

to read the angle via an SPI Read of the READC register. The READC register contains an 8-bit value which is incremented each time the register is read. The controller can verify that the sensor is working properly by checking if the value is updated at every SPI Read.

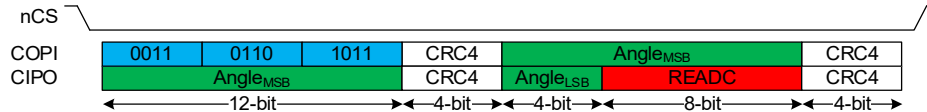


Figure 16: SPI Read Angle

There are several options to perform shorter angle reads for time optimization:

- 16-bit angle read in 20 clock cycles (Figure 17). In this case, though, the 4-bit Angle_LSB is not protected by CRC and an eventual communication error will not be detected.

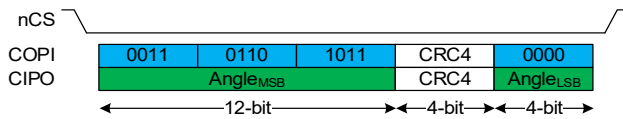


Figure 17: 20 Clock Cycles Read Angle

- 12-bit angle read with CRC4 in 16 clock cycles (Figure 18)

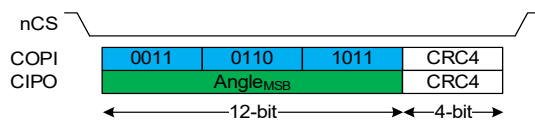


Figure 18: 16 Clock Cycles Read Angle

- 12-bit angle read without CRC4 in 12 clock cycles (Figure 19). In this case too, an eventual communication error will not be detected.

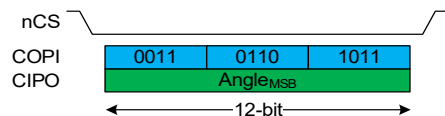


Figure 19: 12 Clock Cycles Read Angle

SPI Error Detection

The MA900B can detect multiple types of errors during the SPI communication. Whenever an SPI error is detected:

- the nFAULT pin is driven low
- the SPI fault flag in the IFAIL register and the IFAIL flag in the STATUS register are set to 1.
- If the SPI CRC Inversion is enabled (CRCINVEN = 1), each CRC sent on CIPO after the error detection is bitwise inverted (~CRC4) until the SPI error flag is cleared.

If CRCINVEN = 1, the CRC inversion is performed also if any other fault flag is raised.

First Frame Invalid OpCode or CRC

An error is detected if the OpCode sent by the controller does not match either a Read (0b0011) or Write (0b1100) operation. The operation is converted to a read of the STATUS register. Figure 20 shows the sensor response when the SPI CRC Inversion is enabled, while Figure 21 when the SPI CRC Inversion is disabled.



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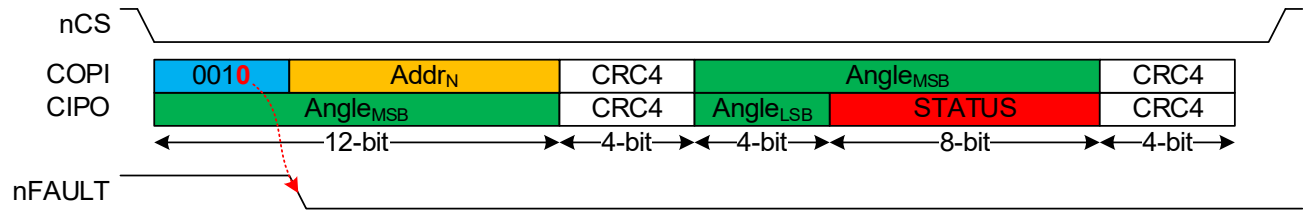


Figure 20: Wrong OpCode Detection, CRCINVEN = 0

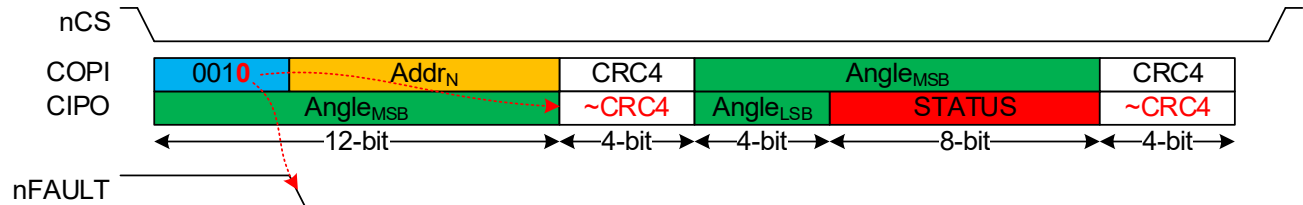


Figure 21: Wrong OpCode Detection, CRCINVEN = 1

If the CRC is enabled, the corruption of the address value Addr_N can also be detected. In this case, the error is detected because the first CRC4 sent by the controller on COPI does not

match the one calculated by the MA900B (Figure 22 and Figure 23). The operation is converted to a read of the STATUS register.

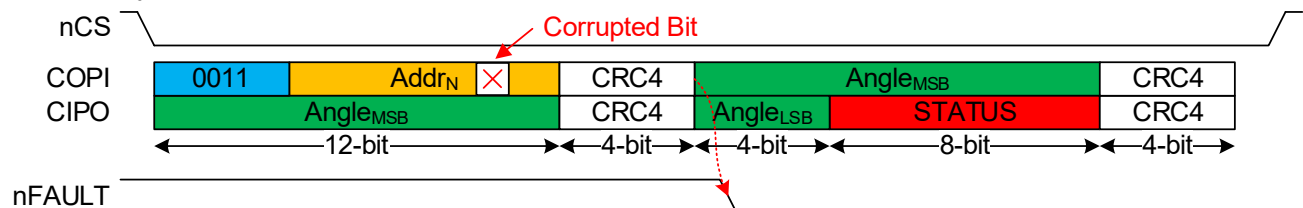


Figure 22: Error Detection After a Valid OpCode (Read), CRCINVEN = 0

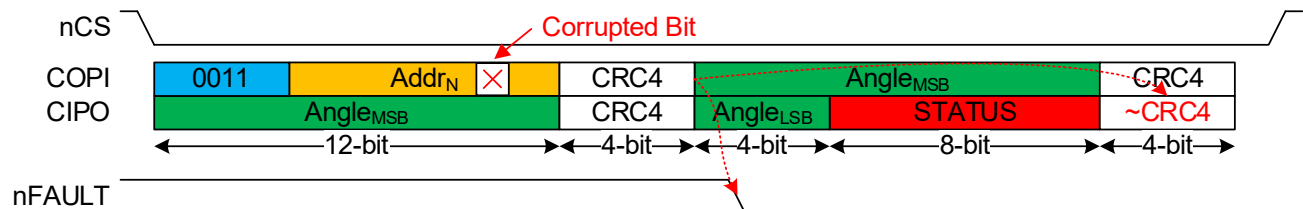


Figure 23: Error Detection After a Valid OpCode (Read), CRCINVEN = 1

Write Error

Data corruption during a Write operation is detected by the MA900B when the CRC is enabled, as the CRC4 sent by the controller after the Data_N does not match the value calculated by the MA900B, see Figure 24.

In this case, the write is not performed. The controller can detect this by checking that the

data returned by the sensor does not match the sent data.

It is recommended to always end the Write operation with 0b111100000000 code to verify that the last write operation was performed correctly.

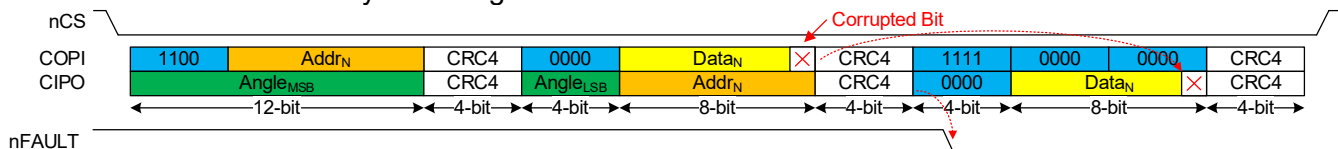


Figure 24: SPI Write Error, CRCINVEN = 0



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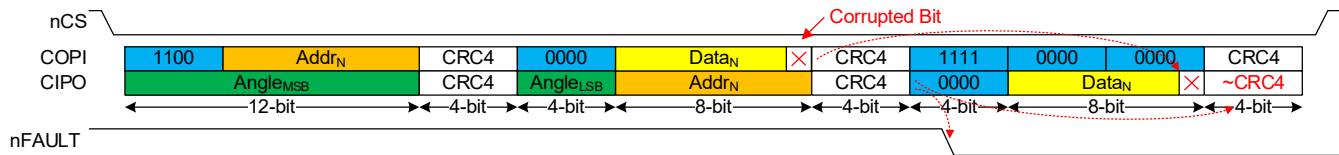


Figure 25: SPI Write Error, CRCINVEN = 1

Handshaking Error

If the SPI Handshaking is enabled (SPIHS = 1) and the controller sends a wrong handshaking, an error is detected.

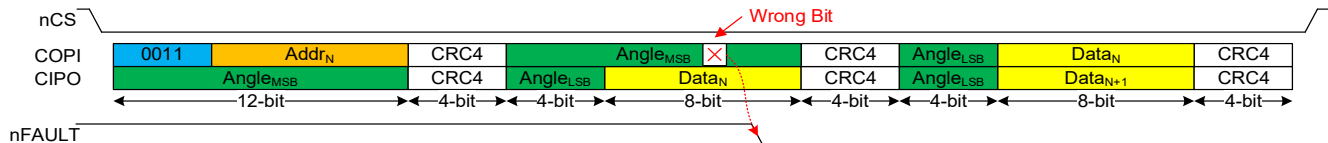


Figure 26: Handshaking Error, CRCINVEN = 0

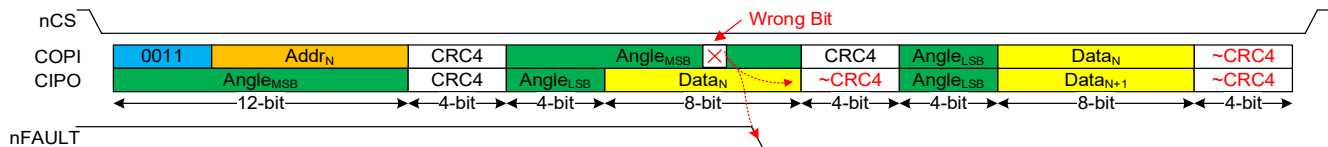


Figure 27: Handshaking Error, CRCINVEN = 1

SPI Pins Configuration

The SPI input pins internal circuit is configurable by the SPIIPULL bit in the IO1 register. The pins can be configured as:

- High-impedance
- Internal pull-up to VDD (nCS)
- Internal pull-down to GND (SCLK, COPI)

When they are configured as high-impedance, it is recommended to externally pull SCLK and COPI to GND and CS to VDD, to avoid any voltage build-up and undesired programming when the SPI is idle.

Figure 28 and Figure 29 show the SPI Input pins internal circuits.

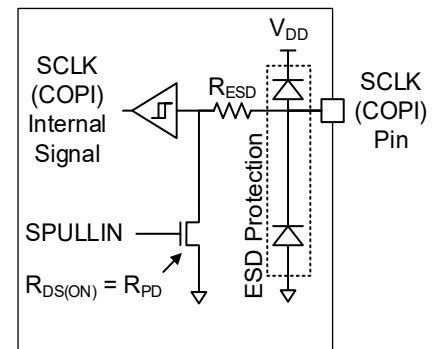


Figure 28: Internal Circuit of SCLK and COPI Pins

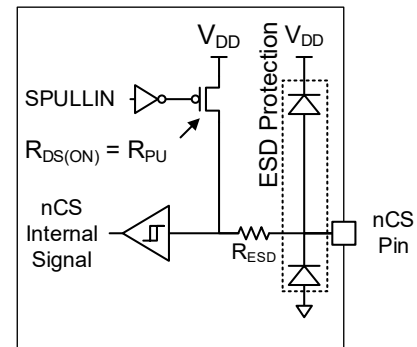


Figure 29: Internal Circuit of nCS Pin



The SPI output pin (CIPO) internal circuit when the SPI is idle is configurable by SPIOPULL bit in the IO1 register. The pin can be configured as:

- High-impedance
- Internal pull-down to GND

The configuration of the pin as high-impedance allows to connect multiple devices on the same SPI bus.

Figure 30 and Figure 31 show the CIPO pin internal circuit.

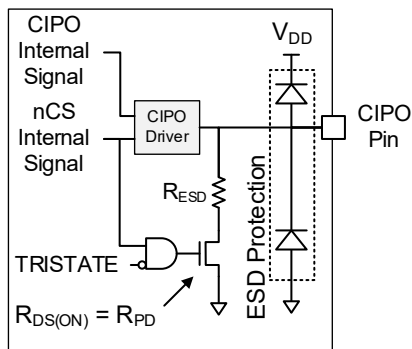


Figure 30: Internal Circuit for SPI CIPO Pin

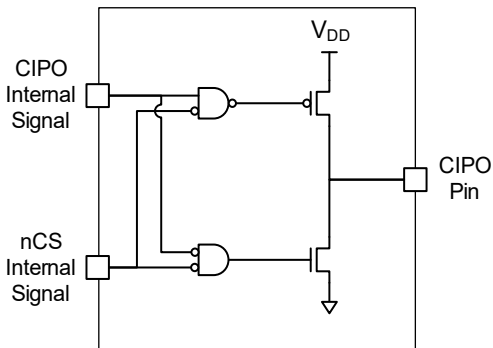


Figure 31: 'CIPO Driver' Block Circuit

SPI Reduced Wiring Mode

The reduced wiring mode (RWM) allows to spare 3 connections on the PCB by swapping the function of the pins 5, 6, 7, used by default for SPI interface, with the function of the pins 8, 9, 10 (Table 5).

The RWM is enabled by the RWM bit in the INTF register. This bit value can only be changed if WRLOCK = 0 and MTPLOCK=0, see the Lock Mechanisms section on page 25 for additional details.

Changing the RWM bit value does not have an immediate effect. The function of the pins affected by the RWM is assigned at start-up.

To enable the RWM the user needs to:

1. Disable the MTP lock and the Write lock.
2. Set RWM = 1 in the INTF register.
3. Perform an MTP Store (see the MTP Memory Operations on page 24).
4. Power cycle the MA900B.

Significant care should be taken when planning to use the RWM, as the operation is potentially irreversible. Once the RWM is enabled, it is no longer possible to program the sensor if the pins 8, 9, 10 are left unconnected on the PCB.

Table 5: Pin Functionality in RWM Mode

RWM	Pin 5	Pin 6	Pin 7	Pin 8	Pin 9	Pin 10
Disabled	COPI	SCLK	CS	IO0	IO1	IO2
Enabled	IO0	IO1	IO2	COPI	SCLK	CS

I²C

The I²C implementation used in MA900B follows the I²C standard for bidirectional, single controller communication. The sensor operates only as a peripheral device and is designed to operate up to the Fast-Mode Plus (FM+) I²C speed category (1Mbit/s).

MA900B I²C communication can be secured by CRC. When the CRC is enabled (CRCEN = 1), an 8-bit CRC code (CRC8) is transmitted by either the controller or the peripheral device after each data byte. See APPENDIX A: CRC Calculation for CRC calculation details.



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Mandatory I²C features for peripheral device defined by NXP UM10204 - I²C-bus specification and user manual are supported.

I²C Sensor Address

The I²C 7-bit Sensor Address (also referred to as Target Address in the NXP I²C-bus specification) is defined by the logic level of the SA0, SA1, SA2 pins at start-up (see Table 6). The sensor address is not updated if the SA0, SA1, SA2 pins value changes after start-up.

Table 6: I²C Sensor Address

I ² C Sensor Address (7-bit)						
1	1	0	0	SA2	SA1	SA0

I²C Operations

The following I²C operations are supported on MA900B:

- Read
- Quick Read
- Write

Each operation can be performed on a single register or multiple consecutive registers.

I²C Operations Common Features

The I²C frames used for Read, Quick Read and Write operations share some common characteristics:

- The communication is organized in 8-bit sized frames.
- At the end of each frame, the communication continues only if the device that was not driving the SDA line during the frame sends an Acknowledge (ACK) bit, by pulling the SDA line to 0. If the SDA line is kept to 1, this

is considered as a Not Acknowledge (NACK) bit, which stops the communication.

- The controller initiates the communication by sending a START condition and terminates it by sending a STOP condition.

I²C Read

An I²C Read operation allows the controller to receive the value of one or multiple consecutive registers. It is formed by the following sequence:

- In the first frame, the controller sends the Sensor Address in the 7 most significant bits, and 0 in the least significant bit.
- The sensor sends the ACK bit if the Sensor Address is recognized.
- In the second frame, the controller sends the 8-bit address of the register to be read.
- After the ACK bit, the controller sends a repeated START condition
- In the third frame, the controller sends the Sensor Address in the 7 most significant bits, and 1 in the least significant bit.
- In the fourth frame, the sensor sends the 8-bit register value (Figure 32).
- To receive the value of the next register, the controller can send an ACK bit and keep sending clock pulses on the SCL line (Figure 33). This operation can be repeated multiple times and is only limited by the number of available registers.
- If CRC is enabled (CRCEN = 1), the sensor sends the CRC8 value after each register value (Figure 34).

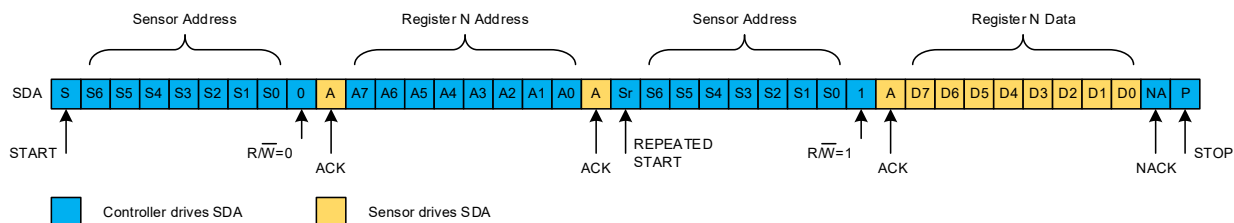


Figure 32: I²C Read Single



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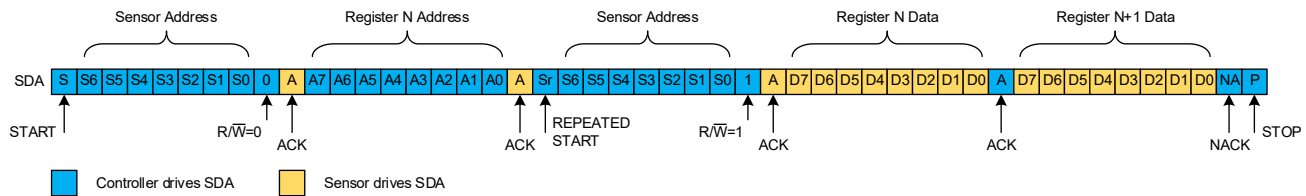


Figure 33: I²C Read Burst

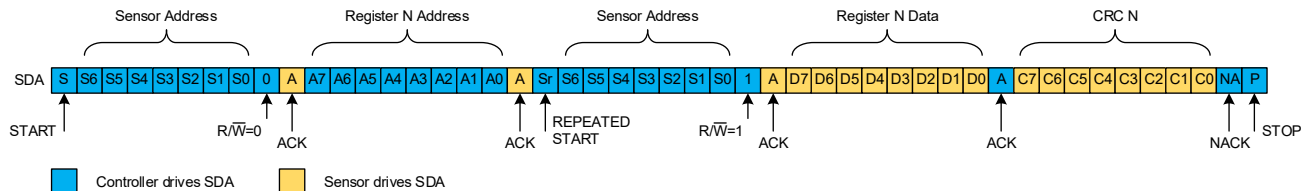


Figure 34: I²C Read Single with CRCEN = 1

I²C Quick Read

The Quick Read operation is a faster option to read the measurement related registers (from ANGLE0 to TEMP1). It is formed by the following sequence:

- In the first frame, the controller sends the Sensor Address in the 7 most significant bits, and 1 in the least significant bit.
- The sensor sends the ACK bit if the Sensor Address is recognized.
- In the second frame, the controller sends the ANGLE0 register value.
- As for the I²C Read case, the controller determines how many register values are sent by the sensor. Figure 35 shows an I²C Quick Read operation to read the full 16-bit angle.
- If CRC is enabled (CRCEN = 1), the sensor sends the CRC8 value after each register value (Figure 36).

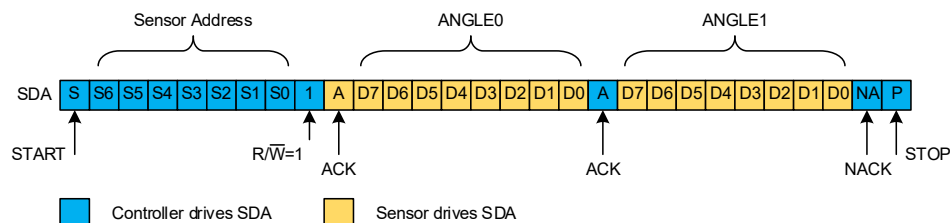


Figure 35: 16-bit angle I²C Quick Read

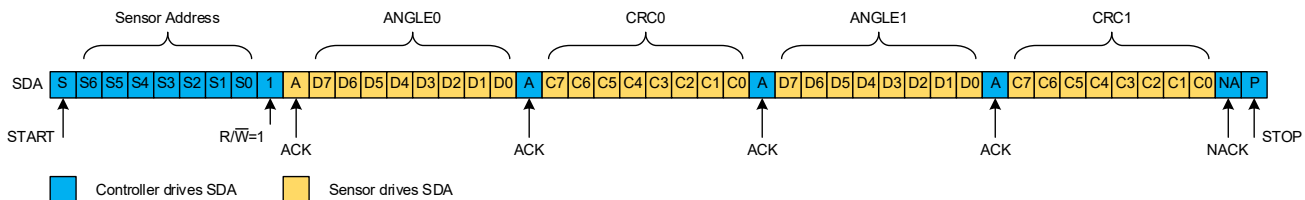


Figure 36: 16-bit angle I²C Quick Read with CRCEN = 1

I²C Write

The I²C Write operation allows the controller to change the value of one or multiple consecutive registers. It is formed by the following sequence:

- In the first frame, the controller sends the Sensor Address in the 7 most significant bits, and 0 in the least significant bit.
- The sensor sends the ACK bit if the Sensor Address is recognized.
- In the second frame, the controller sends the 8-bit address of the register where the new value needs to be written.
- In the third frame, the controller sends the 8-bit value to write in the register (Figure 37).



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- To write in the following register, the controller sends an additional frame with the 8-bit value to write in it (Figure 38).
- If CRC is enabled (CRCEN = 1), the controller must send the CRC after each

value sent (Figure 39). If the controller does not send the CRC or sends a wrong CRC, the I²C Write is ignored and the IFAIL and I2C flags, respectively in the STATUS and IFAIL registers, are set to 1.

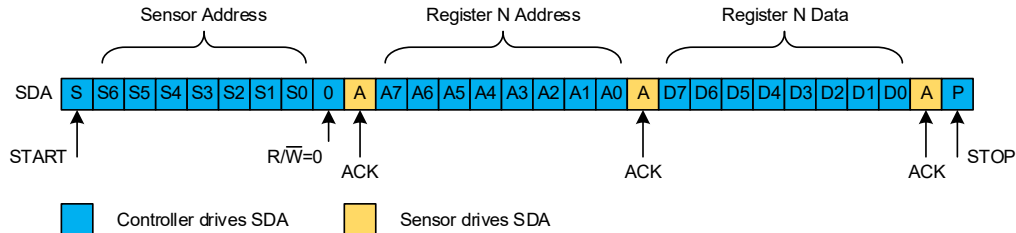


Figure 37: I²C Write Single

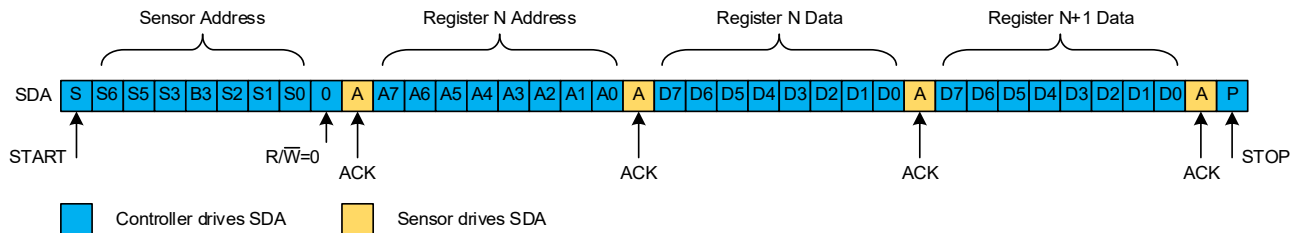


Figure 38: I²C Write Burst

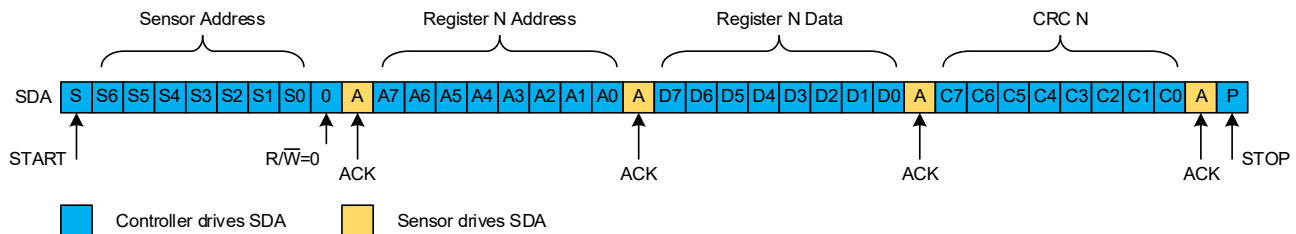


Figure 39: I²C Write Single with CRC

Measurement values latching

By default, the Angle, Speed, Multiturn and Temperature values are latched during the entire I²C transactions, namely between a START and a STOP condition. This ensures all the values read during the I²C transactions are coherent.

MTP Memory Operations

The value of the registers from ZERO0 to PID can be stored in an MTP (Multi-Time Programmable) memory, to keep the sensor settings over successive power cycles. MTP Store/Restore operations can be performed both via SPI and I²C.

MTP memory operations are protected by a lock mechanism to prevent unwanted modification of the MTP memory content. See the Lock

Mechanisms section on page 25 for more information.

MTP Store

The MTP Store operation saves the current value of the registers from ZERO0 to PID in the MTP memory.

This operation is performed by setting to 1 the STORE bit in the MTP register. The STORE bit is automatically cleared as soon as operation starts. The time required to perform an MTP Store is shown in the General Characteristics table.

MTP Restore

The MTP Restore operation to copies the values stored in the MTP memory to the registers from



ZERO0 to PID. This operation is performed by setting the RESTORE bit in the MTP register to 1. The bit is automatically cleared as soon as the operation starts. The time required to perform an MTP Restore is shown in the General Characteristics table.

MTP Flags

During MTP memory operations, the MTPBUSY flag in the STATUS register is set to 1.

When MTPBUSY = 1, the registers from ZERO0 to PID are not accessible: Write operations are discarded and Read operations return 0. Additionally, the IFB flag in the IFAIL register is set and the nFAULT pin is driven low. It is strongly recommended to avoid Write or Read operations on other registers too.

After a successful MTP operation, the MTPBUSY flag is set to 0.

The MTPDONE flag is the logic inversion of the MTPBUSY flag.

The MTP memory is divided into blocks and, during restore operations, an ECC algorithm is used to check for errors in any of the blocks. If an error is detected, there are two possible reactions:

- If the error is correctable, the MTP1 flag in the IFAIL register is raised, the error is corrected and the volatile memory registers corresponding to blocks where the error is detected are updated.
- If the error is uncorrectable, the MTP2 flag in the IFAIL register is raised and the volatile memory registers corresponding to blocks where an uncorrectable error is found are not updated.

Lock Mechanisms

MA900B features two register write-lock options to prevent undesired modification to its registers:

- MTP lock: the MTP register is write-locked. When this register is locked, neither MTP Store nor MTP Restore operations can be executed.
- Write lock: all the registers except the LOCK register are write-locked.

A specific sequence of write operations, either SPI or I²C, in the LOCK register is required to toggle the status of each lock (from lock to unlock or vice versa). The sequence is shown in the LOCK register description on page 48.

The status of the locks can be read in the LOCKST register.

The default/after-restore status of the Write lock is configurable by using the WLOCK bit in the INTF register.

SSI

SSI is a 2-wire synchronous serial interface. The sensor operates as a peripheral to the external SSI controller.

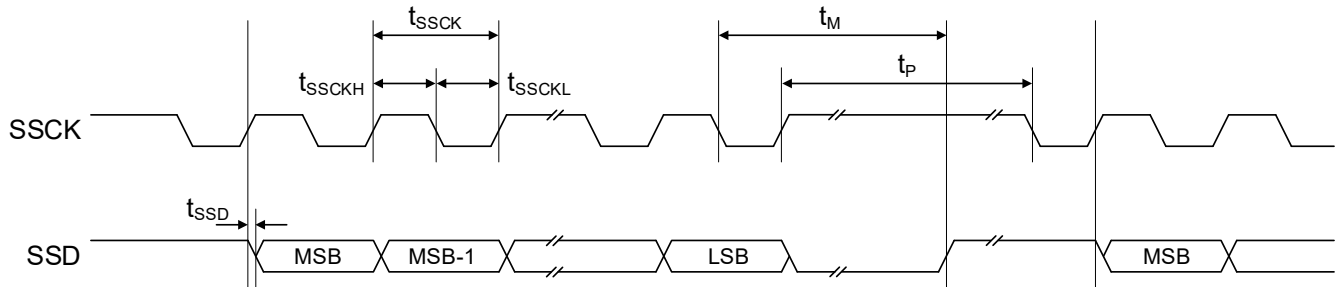
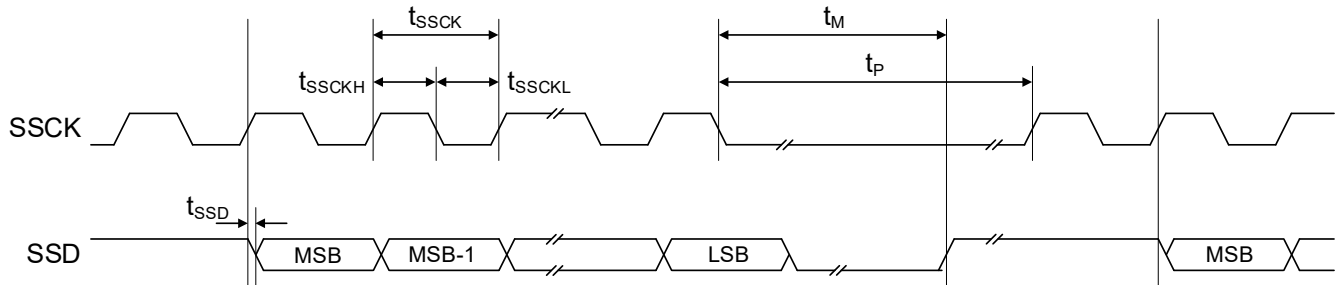
SSI can only be used for angle reading. It is not possible to read or write registers using SSI.

The maximum SSI clock frequency supported by MA900B is 5MHz. The actual data rates depend also on the PCB layout quality and signal trace length.

Two SSI modes are supported:

- In Mode A, SSCK is held high when communication is idle (Figure 40)
- In Mode B, SSCK is held low when communication is idle (Figure 41)

Timing details of SSI communication are shown in Table 7.


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Figure 40: SSI Mode A

Figure 41: SSI Mode B
Table 7: SSI Timings

Parameter	Description	Min	Max	Unit
t_{SSD}	Delay between SCK and SSD edge		50	ns
t_{SSCK}	SSCK period	0.2	20	μ s
t_{SSCKL}	Low level of SCK signal	0.1	10	μ s
t_{SSCKH}	High level of SCK signal	0.1	10	μ s
t_M	Transfer timeout (monoflop time)	25		μ s
t_P	Dead time: SCK high time for next data reading	40		μ s

SSI Read Angle

To read the angle via SSI, the controller must send clock pulses on the SCLK line. The first edge on SCK (falling for Mode A, rising for Mode B) is the trigger event signaling the start of a new transmission. After the first edge detection, the sensor sends, on the SSD line, the angle 16-bit sequence, MSB first. The SSD level is updated on each SCK rising edge and the controller samples it on the following falling edge.

Since the first clock pulse on SCK is used to trigger a new transmission, reading the 16-bit angle requires 17 clock pulses (see Figure 42

and Figure 43). The transmission length can be reduced if a shorter angle data is sufficient for the application. For example, if an application requires only a 12-bit angle value, it is sufficient to send 13 clock pulses on SCK.

When a trigger event is detected, the current angle data is held in the output buffer until the transfer timeout t_M has passed from the last SCK falling edge.

Consecutive angle readings need to be separated by a minimum time delay t_P see (Figure 44).



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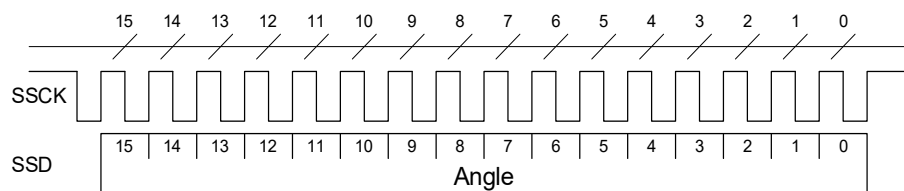


Figure 42: SSI Mode A Angle Read

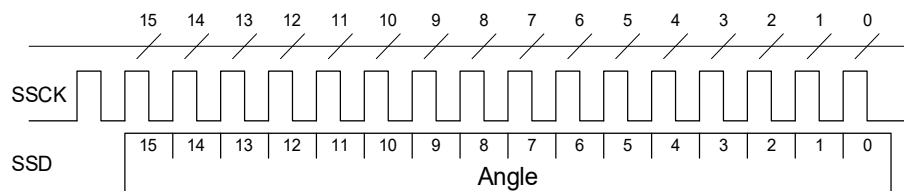


Figure 43: SSI Mode B Angle Read

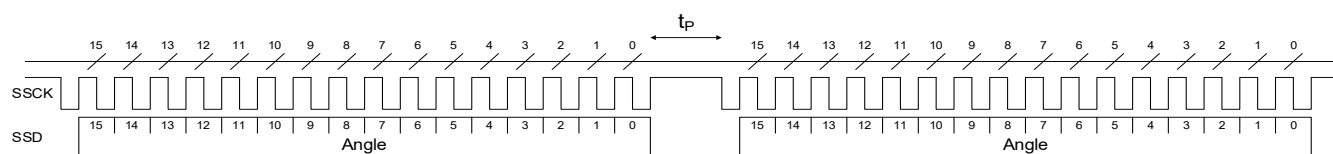


Figure 44: Two consecutive 16-bit SSI Angle Reads (Mode A)

SSI Parity Check

SSI data integrity can be validated via a parity bit.

The parity sign can be configured via the SSIP bit in the SSI register.

The parity bit is always sent after the angle LSB. To read the parity bit, then, the controller needs to send 18 clock pulses (Figure 45).

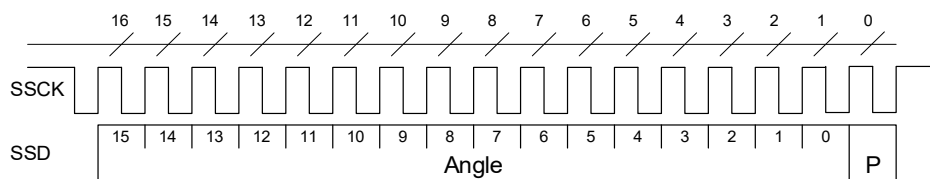


Figure 45: SSI Angle Read with Parity Check

SENT

The SENT (Single Edge Nibble Transmission) protocol is a one-wire unidirectional protocol. The sensor is continuously transmitting information and does not need any action from the receiving device. MA900B supports SENT interface based on *SAE J2716 APR2016* Specification.

A SENT message consists of a sequence of pulses. The pulse width, the time between two consecutive falling edges, is always a multiple of a fixed time unit, the tick.

Figure 46 shows the SENT message pulses sequence:

1. A Synchronization/Calibration pulse with a fixed 56 ticks length

2. A 4-bit Status/Serial Communication nibble pulse
3. Up to six 4-bit Data nibble pulses. The information transmitted on these pulses are also referred to as 'fast channels'. The number fast channels and their content depend on the configured data frame format (see the Data Frame Format section)
4. A 4-bit CRC nibble pulse
5. A Pause pulse. This pulse is optional, see the Pause Pulse section on page 30 for more information.



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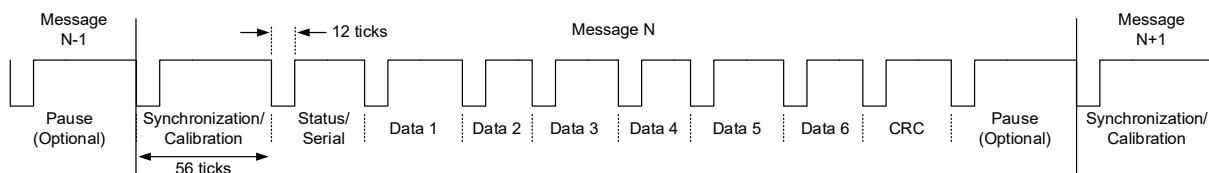


Figure 46: Sent Message Pulses Sequence

All pulses have a minimum width of 12 ticks. For each pulse, the signal is low for at least 5 ticks.

The nibble value is the pulse width in ticks beyond the minimum. For example, if the nibble value is 10, its width is $12 + 10 = 22$ ticks (see Figure 47).

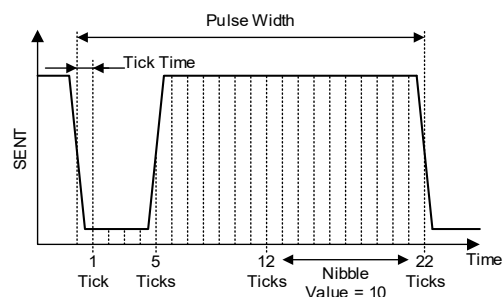


Figure 47: Pulse Width and Nibble Value

The SENT configurable options supported by MA900B are listed in the next sections. For additional information about the SENT protocol, refer directly to the SAE J2716 APR2016 Specification.

Data Frame Format

MA900B supports all the data frame formats listed in the SAE J2716 specification (Table 8). For additional details on the formats, refer to Appendix H of the specification.

The data frame format is selected via SENTFORMAT bit in the SENT0 register.

The type of data transmitted on Channel 2 can be configured by the SENTCH2 bit in the SENT1 register.

Table 8: SENT Data Frame Formats

Frame Format	Description	Data Pulses
H.1	Two 12-bit fast channels	6
H.2	One 12-bit fast channel	3
H.3	High-speed with one 12-bit fast channel (3 bits per nibble)	4
H.4	Secure sensor with 12-bit fast channel 1 and secure sensor information on fast channel 2	6
H.5	Single sensor with 12-bit fast channel 1 and zero value on fast channel 2	6
H.6	Two fast channels with 14-bit fast channel 1 and 10-bit fast channel 2	6
H.7	Two fast channels with 16-bit fast channel 1 and 8-bit fast channel 2	6

Status Nibble

The information contained in the Status/Serial Communication nibble is shown in Table 9.

Table 9: Status Nibble Bits

Status Nibble Bit	Meaning
0	Fault
1	Reserved
2	Serial Data Message Bits
3	

Serial Message

MA900B supports the transmission of Serial Message (see sections 5.2.4.1 and 5.4.2.2 of SAEJ2716 specification). This feature is configurable via the SENTSERIAL bits in the SENT1 register.

The sequences of messages sent for Short and Enhanced serial message formats are shown in Table 10, Table 11 and Table 12. The Error Codes for each Serial Message format are shown in Table 13. See the AFAIL, DFALL and


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IFAIL registers descriptions for additional details
on the error flags meaning.

Table 10: Short Serial Message Sequence

Message Index	Message ID (4-bit)	Message Data (8-bit)	Description
0	0x1	Error	Error message (J2716 D.5)
1	0x6	0x04	SENT Standard Revision - Revision 4
2	0x1	Error	Error message (J2716 D.5)
3	0x5	0x00	Manufacturer Code - Not Specified
4	0x1	Error	Error message (J2716 D.5)
5	0x3	0x50	Channel 1/2 Sensor type (J2716 D.4) - Not Specified Position Sensors
6	0x1	Error	Error message (J2716 D.5)
7	0x0	Temp	Temperature (Signed 8-bit, 2° per LSB)
8	0x1	Error	Error message (J2716 D.5)
9	0xB	0x00	Zero
10	0x1	Error	Error message (J2716 D.5)
11	0xC	0x00	Zero
12	0x1	Error	Error message (J2716 D.5)
13	0xD	0x00	Zero
14	0x1	Error	Error message (J2716 D.5)
15	0xE	0x00	Zero

Table 11: Enhanced Serial Message (8-bit ID/12-bit data) Sequence

Message Index	Message ID (8-bit)	Message Data (12-bit)	Description
0	0x01	Error	Error message (J2716 D.5)
1	0x06	0x004	SENT Standard Revision - Revision 4
2	0x01	Error	Error message (J2716 D.5)
3	0x05	0x000	Manufacturer Code - Not Specified
4	0x01	Error	Error message (J2716 D.5)
5	0x03	0x050	Channel 1/2 Sensor type (J2716 D.4) - Not Specified Position Sensors
6	0x01	Error	Error message (J2716 D.5)
7	0x23	Temp	Temperature (12-bit signed, 0.125° per LSB)
8	0x01	Error	Error message (J2716 D.5)
9	0x29	0x000	Zero
10	0x01	Error	Error message (J2716 D.5)
11	0x2A	0x000	Zero
12	0x01	Error	Error message (J2716 D.5)
13	0x2B	0x000	Zero
14	0x01	Error	Error message (J2716 D.5)
15	0x2C	0x000	Zero



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PRELIMINARY SPECIFICATION SUBJECT TO CHANGE**Table 12: Enhanced Serial Message (4-bit ID/16-bit data) Sequence**

Message Index	Message ID (4-bit)	Message Data (16-bit)	Description
0	0x1	Error	Error message (J2716 D.5)
1	0x6	0x0004	SENT Standard Revision - Revision 4
2	0x1	Error	Error message (J2716 D.5)
3	0x5	0x0000	Manufacturer Code - Not Specified
4	0x1	Error	Error message (J2716 D.5)
5	0x3	0x0050	Channel 1/2 Sensor type (J2716 D.4) - Not Specified Position Sensors
6	0x1	Error	Error message (J2716 D.5)
7	0x2	Temp	Temperature (16-bit signed, 0.125° per LSB)
8	0x1	Error	Error message (J2716 D.5)
9	0xB	0x0000	Zero
10	0x1	Error	Error message (J2716 D.5)
11	0xC	0x0000	Zero
12	0x1	Error	Error message (J2716 D.5)
13	0xD	0x0000	Zero
14	0x1	Error	Error message (J2716 D.5)
15	0xE	0x0000	Zero

Table 13: Serial Message Error Code (16-bit, 12-bit and 8-bit)

16-bit Error	12-bit Error	8-bit Error	Error Flag
0x0000	0x000	0x00	No error
0x8001	0x801	0x01	MGL
0x8002	0x802	0x02	MGH
0x8004	0x804	0x04	ASUPPLY
0x8008			DSUPPLY
0x8080	0x810	0x10	OOL
0x9000	0x840	0x40	REGCRC
0xA000	0x880	0x80	MTP1
0xC000			MTP2

CRC

MA900B supports both recommended and legacy CRC implementations described in Section 5.4.2 of the SAE J2716 specification.

The CRC type can be configured via the SENTCRC bit in the SENT1 register.

Pause Pulse

A pause pulse can be added to the SENT frame. The purpose of the pause pulse is to have a constant frame length. The enabling and the length of the pause pulse are configured by the SENTPAUSE bit in the SENT1 register.

Tick Length

The length of a single SENT tick is configurable through the SENTTICK bit in the SENT0 register.

SPC Protocol

SPC protocol is supported by MA900B. It turns SENT into a half-duplex synchronous communication, allowing the controller to send a trigger to request data transmission from the sensor. It is enabled by the SPCMODE bit in the SENT0 register.

The SPC pulses sequence is shown in Figure 48. The communication is triggered by the controller by pulling the SENT line low for a time t_{MT} , which



needs to be of at least 2 ticks. The sensor response time t_{RES} is fixed to 90 ticks.

An End pulse with a length of 12 ticks is added at the end of the frame.

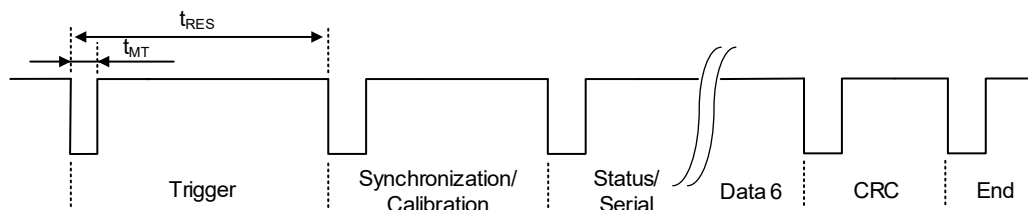


Figure 48: SPC Pulses Sequence

SENT Fault Detection

The SENT output is monitored to detect if the IO pad is shorted to VDD or GND (as specified in SAE J2716, section 6.6).

The monitoring is not done continuously. The SENT pad voltage is checked only at ticks 4 and 11 of every pulse, see Figure 49.

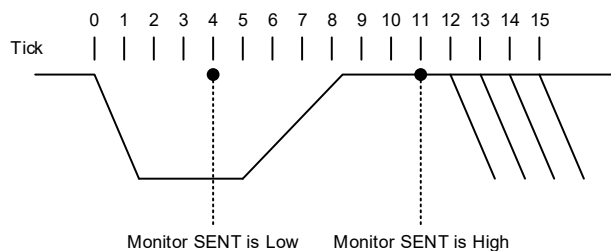


Figure 49: SENT Pad Fault Detection

If a short fault is detected, the SENT interface is reset. Upon removal of the fault, the SENT interface resumes normal operation.

When the SENT pad fault is detected, the SENT flag in the IFAIL register is set to 1.

Falling Edge Slew Rate

SENT output falling edge slew rate can be configured to reduce EMC (See section 6.3 of the J2716 specification), via the SENT_SR bit in the SENT2 register.

The rising edge slew rate can be adjusted by changing the external pull-up value. The recommended value is between 10kΩ and 55kΩ.



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PRELIMINARY SPECIFICATION SUBJECT TO CHANGE**REGISTER MAP****Table 14: Register Map**

Addr. (Dec)	Addr. (Hex)	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
0	00	ZERO[7:0]							
1	01	ZERO[15:8]							
2	02	HYST							
3	03	PPT[7:0]							
4	04	DAZ	ILIP				PPT[10:8]		
5	05	RESERVED					NPP		
6	06	MGLT				MGHT			
7	07	RESERVED	PBI_TIME			PBI_EN	RESERVED		
8	08	SPCMODE	SENTFORMAT			SENTTICK			
9	09	SENTCRC	RESERVED	SENTCH2		SENTPAUSE		SENTSERIAL	
10	0A	RESERVED					SENT_SR		
11	0B	RESERVED						PWMM	PWMF
12	0C	RESERVED							SSIP
15	0F	RWM	I2CEN	RESERVED	WLOCK	RESERVED	SPIHS	CRCINVEN	CRCEN
16	10	RESERVED							RD
17	11	RESERVED				FW			
18	12	SFEN	LAGCEN	RESERVED		FWS			
20	14	RESERVED				IOM			
21	15	RESERVED		SPIIPULL	SPIOPULL	DRV		OD012	OD3456
31	1F	PRODUCTID				RESERVED			
96	60	ANGLE [7:0]							
97	61	ANGLE [15:8]							
98	62	SPEED [7:0]							
99	63	SPEED [15:8]							
100	64	MULTITURN [7:0]							
101	65	MULTITURN [15:8]							
102	66	TEMP [7:0]							
103	67	TEMP [15:8]							
106	6A	GPIODIR	GPIO						
107	6B	READ COUNTER							
108	6C	MTPBUSY	RESERVED	INITBUSY	RESERVED		IFAIL	DFAIL	AFAIL
109	6D	RESERVED	OOL	RESERVED		DSUPPLY	ASUPPLY	MGH	MGL
110	6E	RESERVED			REGCRC	MTP2	MTP1	RESERVED	
111	6F	IFB	RESERVED				SENT	I2C	SPI
115	73	RESERVED	I2CSA						
119	77	RESERVED				ET_SHIFT			ET
120	78	MTPDONE	RESERVED	STORE	RESTORE	RESERVED			
126	7E	RESERVED					MTPLOCK	RESERVED	WRLOCK
127	7F	LOCK							



REGISTERS DESCRIPTION

The symbols used to indicate the bits access type are summarized in Table 15.

Table 15: Bits Access Type Symbols Definition

Symbol	Access Type
R	Read-Only
R/W	Read/Write
R/W1C	Read/Write 1 to Clear

ZERO0 (0x00)

The ZERO0 register is used to configure the sensor zero, see the Zero and Rotation Direction section on page 12 for more details.

Bits	Access	Bit Name	Default	Description
7:0	R/W	ZERO[7:0]	0	8 least significant bits (LSB) of the sensor zero. See the ZERO1 register description for the conversion formula to degrees.

ZERO1 (0x01)

The ZERO1 register is used to configure the sensor zero, see the Zero and Rotation Direction section on page 12 for more details.

Bits	Access	Bit Name	Default	Description
7:0	R/W	ZERO[15:8]	0	8 most significant bits (MSB) of the sensor zero. The angle α_{out} output by the MA900B is calculated as shown in Equation (2): $\alpha_{out} = \alpha_{raw} - \alpha_0 \quad (2)$ α_{raw} is the raw angle provided by the MA900B digital filter and α_0 is the sensor zero. Equation (3) shows the relationship between α_0 and ZERO[15:0]: $\alpha_0 = 360 \times \frac{ZERO[15:0]}{2^{16}} \quad (3)$ Equation (4) shows a calculation example to set the zero position to 20°: $ZERO[15:0] = \frac{20}{360} \times 2^{16} = 3641 \quad (4)$


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HYST (0x02)

The HYST register is used to configure ABZ and UVW interfaces hysteresis. See the ABZ Hysteresis section on page 51 and the UVW Hysteresis section on page 52 for more details.

Bits	Access	Bit Name	Default	Description
7:0	R/W	HYST	219	Sets the Hysteresis for ABZ and UVW interfaces. If $\text{HYST}[6:0] = 0$, the hysteresis H (in degrees) is set to its minimum value: $H_{\text{MIN}} = \frac{360}{2^{15}} \quad (5)$ If $\text{HYST}[6:0] \neq 0$ and $\text{HYST}[7] = 1$, the hysteresis depends on the FW bits value. If $\text{FW} = 0$: $H = \frac{360}{2^{15}} \times 2^{\frac{\text{HYST}[6:0] - 1}{12}} \quad (6)$ If $\text{FW} \neq 0$: $H = \frac{360}{2^{15}} \times 2^{\frac{\text{HYST}[6:0] + 5 - 6 \times \text{FW}}{12}} \quad (7)$ In the latter case, H is always lower bounded by H_{MIN}. If $\text{HYST}[6:0] \neq 0$ and $\text{HYST}[7] = 0$, the hysteresis is: $H = \frac{360}{2^{15}} \times 2^{\frac{\text{HYST}[6:0] - 1}{12}} \quad (8)$

ABZ0 (0x03)

The ABZ0 register is used to configure ABZ output Pulses Per Turn (PPT). See the ABZ Incremental Encoder Output section on page 50 for more details.

Bits	Access	Bit Name	Default	Description
7:0	R/W	PPT[7:0]	255	Bits 7 to 0 of the PPT setting. See the PPT[10:8] bits description for a calculation example.

ABZ1 (0x04)

The ABZ1 register is used to configure ABZ output PPT, Z index length and position, and to select between ABZ or DAZ interface. See the ABZ Incremental Encoder Output section on page 50 and the DAZ Interface section on page 51 for more details.

Bits	Access	Bit Name	Default	Description
7	R/W	DAZ	0	Converts the ABZ interface to DAZ (D = Direction, A = pulse, Z = index). 0: ABZ 1: DAZ
6:3	R/W	ILIP	0	Configures position and length of the Z pulse. See Figure 54 on page 51 for programming examples.
2:0	R/W	PPT[10:8]	3	Bits 10 to 8 of the PPT setting. The number of pulses per turn of the A and B signals is $\text{PPT} + 1$. Numerical example: - Desired number of Pulses per Turn = 120 $\text{PPT} = 120 - 1 = 119$ (decimal) = 0b00001110111 (binary) $\text{PPT}[10:8] = 0b000$ $\text{PPT}[7:0] = 0b01110111$


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UVW (0x05)

The UVW register is used to configure the number of pole pairs of the UVW output. See the UVW section on page 51 for more details.

Bits	Access	Bit Name	Default	Description
7:3	-	RESERVED	-	Reserved.
2:0	R/W	NPP	0	Configures the number of pole pairs emulated by the UVW interface per magnetic period. 0: 1 pole pair 1: 2 pole pairs 2: 3 pole pairs 3: 4 pole pairs 4: 5 pole pairs 6: 7 pole pairs 7: 8 pole pairs

BLEVEL (0x06)

The BLEVEL register is used to configure the MagHigh (MGH), MagLow (MGL) and Push Button Interface (PBI) thresholds. See the Magnetic Field Detection Flags section on page 53 for more details.

Bits	Access	Bit Name	Default	Description
7:4	R/W	MGLT	0	If PBI_EN = 0, configures the magnetic field threshold for MGLT rising and falling edge. See Equations (9) and (10) for the conversion formulas from LSB to mT. $\text{MGL Rising Edge Threshold [mT]} = 7 \times (\text{MGLT} + 1) \quad (9)$ $\text{MGL Falling Edge Threshold [mT]} = 3 + 7 \times (\text{MGLT} + 1) \quad (10)$ If PBI_EN = 1, configures the PBI Falling Edge Threshold. MGLT = 0 should not be used. $\text{PBI Falling Edge Threshold [mT]} = -2 - 0.5 \times \text{MGLT} \quad (11)$
3:0	R/W	MGHT	15	If PBI_EN = 0, configures the magnetic field threshold for MGHT rising and falling edge. See Equations (12) and (13) for the conversion formulas from LSB to mT. $\text{MGH Rising Edge Threshold [mT]} = 7 \times (\text{MGHT} + 1) \quad (12)$ $\text{MGH Falling Edge Threshold [mT]} = 7 \times (\text{MGHT} + 1) - 3 \quad (13)$ If PBI_EN = 1, configures the PBI Rising Edge Threshold. MGHT = 0 should not be used. $\text{PBI Rising Edge Threshold [mT]} = 2 + 0.5 \times \text{MGLT} \quad (14)$


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PBI (0x07)

The PBI register is used to select between PBI or MGH/MGL interface and to configure the PBI time. See the PBI section on page 54 for more details.

Bits	Access	Bit Name	Default	Description
7	-	RESERVED	-	Reserved.
6:4	R/W	PBI_TIME	7	Sets the delay between PBI rising edge and filter reset for PBI. 0: 2 ms 1: 4 ms 2: 8 ms 3: 16 ms 4: 32 ms 5: 65 ms 6: 131 ms 7: 262 ms
3	R/W	PBI_EN	1	Enables the Push Button Interface. 0: Disabled 1: Enabled
2:0	-	RESERVED	-	Reserved.

SENT0 (0x08)

The SENT0 register is used to enable SPC protocol and to configure SENT interface Data Frame format and tick length. See the SENT section on page 27 for more details.

Bits	Access	Bit Name	Default	Description
7	R/W	SPCMODE	0	Enables SPC protocol. 0: Disabled 1: Enabled
6:4	R/W	SENTFORMAT	0	Selects the SENT Data Frame format. 0: H.1 1: H.2 2: H.3 3: H.4 4: H.5 5: H.6 6: H.7 7: Reserved


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3:0	R/W	SENTTICK	7	Selects the SENT tick length. 0: 0 (SENT stopped) 1: 0.25 μ s 2: 0.5 μ s 3: 1 μ s 4: 1.5 μ s 5: 2 μ s 6: 2.5 μ s 7: 3 μ s 8: 3.5 μ s 9: 4 μ s 10: 5 μ s 11: 6 μ s 12: 12 μ s 13: 24 μ s 14: Reserved 15: Reserved Tick lengths below 3 μ s are not included in the SAE J2716 specification.
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SENT1 (0x09)

The SENT1 register is used to configure SENT interface CRC, Channel 2 data, Pause pulse and Serial Data Message format. See the SENT section on page 27 for more details.

Bits	Access	Bit Name	Default	Description
7	R/W	SENTCRC	0	Selects the SENT CRC implementation. 0: Recommended 1: Legacy
6	-	RESERVED	-	Reserved.
5:4	R/W	SENTCH2	0	Selects the information sent on Fast Channel 2. 0: Speed 1: Multiturn 2: Reserved 3: Reserved
3:2	R/W	SENTPAUSE	0	Enables the Pause pulse and selects the frame length. 0: Pause pulse disabled 1: Pause pulse enabled, frame length is 282 ticks 2: Pause pulse enabled, frame length is 400 ticks 3: Reserved
1:0	R/W	SENTSERIAL	0	Selects the SENT Serial Data message format. 0: No Serial Message 1: Short Serial Message 2: Enhanced Serial Message with 8-bit ID / 12-bit data 3: Enhanced Serial Message with 4-bit ID / 16-bit data

SENT2 (0x0A)

The SENT2 register is used to configure SENT interface falling edge slew rate. See the SENT section on page 27 for more details.

Bits	Access	Bit Name	Default	Description
7:3	-	RESERVED	-	Reserved.



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2:0	R/W	SENT_SR	0	Configures the SENT falling edge slew rate 0: 1 V/μs 1: 2 V/μs 2: 4 V/μs 3: 8 V/μs 4: 16 V/μs 5: 32 V/μs 6: 64 V/μs 7: 128 V/μs
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PWM (0x0B)

The PWM register is used to configure the PWM error transmission and frequency. See the PWM Output section on page 52 for more details.

Bits	Access	Bit Name	Default	Description
7:2	-	RESERVED	-	Reserved.
1	R/W	PWMM	0	Enables error transmission for PWM output. 0: Disabled 1: Enabled
0	R/W	PWMF	0	Configures the frequency of PWM output. 0: 1 kHz 1: 250 Hz

SSI (0x0C)

The SSI register is used to configure the SSI parity sign. See the SSI Parity Check section on page 27 for more details.

Bits	Access	Bit Name	Default	Description
7:1	-	RESERVED	-	Reserved.
0	R/W	SSIP	0	Set SSI parity sign. 0: Even 1: Odd

INTF (0x0F)

The INTF register is used to configure SPI and I²C interfaces. See the SPI section on page 15 and I²C section on page 21 for more details.

Bits	Access	Bit Name	Default	Description
7	R/W	RWM	0	Enables Reduced Wiring Mode (RWM), by switching the pins normally used by SPI to a different functionality. 0: Disabled 1: Enabled See the SPI Reduced Wiring Mode section on page 21 for more details.
6	R	I2CEN	See description	Indicates the serial interface enabled on the device. 0: SPI 1: I ² C
5	-	RESERVED	-	Reserved.


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4	R/W	WLOCK	0	Sets the status of the Write lock (in the LOCKST register) at start-up. 0: WRLOCK = 0 1: WRLOCK = 1
3	-	RESERVED	-	Reserved.
2	R/W	SPIHS	0	Enables SPI Handshaking. 0: Disabled 1: Enabled
1	R/W	CRCINVEN	0	Enables SPI CRC Inversion on error detection. 0: Disabled 1: Enabled
0	R/W	CRCEN	0	Enables CRC for SPI and I ² C interfaces. 0: Disabled 1: Enabled See the APPENDIX A: CRC Calculation on page 62 for more information.

DIR (0x10)

The DIR register is used to configure the sensor rotation direction. See the Zero and Rotation Direction section on page 12 for more details.

Bits	Access	Bit Name	Default	Description
7:1	-	RESERVED	-	Reserved.
0	R/W	RD	0	Configures the sensor rotation direction. 0: Clockwise 1: Counterclockwise

FILT (0x11)

The FILT register is used to configure the position output digital filter. See the Position Output Filter section on page 49 for more details.

Bits	Access	Bit Name	Default	Description
7:4	-	RESERVED	-	Reserved.
3:0	R/W	FW	8	Each FW value corresponds to a specific time constant τ , resolution and cutoff frequency. See Table 16 on page 49 for more details.


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SPEEDFILT (0x12)

The SPEED register is used to configure the position and speed output digital filters. See the Position Output Filter section on page 49 and the Speed Output Filter section on page 49 for more details.

Bits	Access	Bit Name	Default	Description
7	R/W	SFEN	1	Enables the use of the speed measurement by the position filter to obtain constant lag in the position measurement. 0: Disabled 1: Enabled (recommended) Contact MPS in case more information is required about the sensor performance outside of the recommended configuration.
6	R/W	LACEN	1	Enables the enhancement of position filter lag compensation to obtain zero lag. 0: Disabled 1: Enabled (recommended) Contact MPS in case more information is required about the sensor performance outside of the recommended configuration.
5:4	-	RESERVED	-	Reserved.
3:0	R/W	FWS	8	Configures the cutoff frequency of the speed output filter, see Table 17 on page 50 for more details.

IO0 (0x14)

The IO0 register is used to configure the function mapping of pins IO0 to IO6. See the IO Pins Function Selection section on page 50 for more details.

Bits	Access	Bit Name	Default	Description
7:4	-	RESERVED	-	Reserved.
3:0	R/W	IOM	0	Configures the function assigned to the IO0 to IO6 pins. See Table 18 on page 50 for the functions corresponding to each pin for each IOM value.

IO1 (0x15)

The IO1 register is used to configure the digital pins.

Bits	Access	Bit Name	Default	Description
7:6	-	RESERVED	-	Reserved.
5	R/W	SPIIPULL	0	Selects the SPI input pins configuration. 0: High-Z 1: Pull-up for CS, Pull-down for SCLK and COPI
4	R/W	SPIOPULL	0	Selects the CIPO pin configuration when the SPI interface is idle (nCS = 1): 0: High-Z 1: Pull-down
3:2	R/W	DRV	0	Configures the drive strength for pins IO0 to IO6. 0: 4 mA 1: 8 mA 2: 12 mA 3: 16 mA


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1	R/W	OD012	0	Selects the configuration of the IO0, IO1 and IO2 pins in GPIO output mode (IOM=15 and GPIODIR=0). 0: Push-pull 1: Open Drain
0	R/W	OD3456	0	Selects the configuration of the IO3, IO4, IO5 and IO6 pins in GPIO output mode (IOM=15 and GPIODIR=0). 0: Push-pull 1: Open Drain

PID (0x1F)

The PID register contains the MA900B Product ID.

Bits	Access	Bit Name	Default	Description
7:4	R	PRODUCTID	7	4-bit MA900B Product ID.
3:0	-	RESERVED	-	Reserved.

ANGLE0 (0x60)

The ANGLE0 register contains the 8 LSBs of the angle measurement.

Bits	Access	Bit Name	Default	Description
7:0	R	ANGLE[7:0]	-	8 LSB of the angle measurement. See the ANGLE1 register description for the conversion formula to degrees.

ANGLE1 (0x61)

The ANGLE1 register contains the 8 MSB of the angle measurement.

Bits	Access	Bit Name	Default	Description
7:0	R	ANGLE[15:8]	-	8 MSB of the angle measurement. The raw angle output can be converted to degrees using Equation (15): $\alpha_{out} = 360 \times \frac{ANGLE[15:0]}{2^{16}} \quad (15)$

SPEED0 (0x62)

The SPEED0 register contains the 8 LSB of the speed measurement.

Bits	Access	Bit Name	Default	Description
7:0	R	SPEED[7:0]	-	8 LSB of the speed measurement. See the SPEED1 register description for the conversion formula to rpm.


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SPEED1 (0x63)

The SPEED1 register contains the 8 MSB of the speed measurement.

Bits	Access	Bit Name	Default	Description
7:0	R	SPEED[15:8]	-	8 MSB of the speed measurement. The speed value is stored as a signed 16-bit integer, using two's complement representation. The speed value in rpm ($speed_{RPM}$) is calculated from the speed decimal value ($speed_{DEC}$) as shown in Equation (16): $speed_{RPM} = speed_{DEC} \times 3.576 \quad (16)$ Equations (17) and (18) show two numerical examples: $SPEED[15:0] = 0x03E8$ $speed_{RPM} = 1000 \times 3.576 = 3576 \quad (17)$ $SPEED[15:0] = 0xFC18$ $speed_{RPM} = -1000 \times 3.576 = -3576 \quad (18)$ The speed measurement range is [-117187, 117183] rpm.

MTV0 (0x64)

The MTV0 register contains the 8 LSB of the multiturn measurement.

Bits	Access	Bit Name	Default	Description
7:0	R/W	MTV[7:0]	0	8 LSB of the multiturn measurement. See the MTV1 register description for more details.

MTV1 (0x65)

The MTV1 register contains the 8 MSB of the multiturn measurement.

Bits	Access	Bit Name	Default	Description
7:0	R/W	MTV[15:8]	0	8 MSB of the multiturn measurement. The multiturn value is stored as a signed 16-bit integer, using two's complement representation. The measurement range is [-32768, 32767] turns. The multiturn value can be shifted by an offset by writing into the MTV0 and MTV1 registers. Equation (19) shows how the MTV value is calculated: $MTV = MTV_{MEAS} + MTV_{OFFSET} \quad (19)$ MTV_{MEAS} is the number of turns measured by the sensor and MTV_{OFFSET} is the user-configured offset. Both MTV_{MEAS} and MTV_{OFFSET} are set to 0 at power-up.


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TEMP0 (0x66)

The TEMP0 register contains the 8 LSB of the temperature measurement.

Bits	Access	Bit Name	Default	Description
7:0	R	TEMP[7:0]	-	8 LSB of the temperature measurement. See the TEMP1 register description for the conversion formula to °C.

TEMP1 (0x67)

The TEMP1 register contains the 8 MSB of the temperature measurement.

Bits	Access	Bit Name	Default	Description
7:0	R	TEMP[15:8]	-	8 MSB of the temperature measurement. The speed value is stored as a signed 16-bit integer, using two's complement representation. The temperature value in °C ($temp^{\circ}C$) is calculated from the temperature decimal value ($temp_{DEC}$) as shown in Equation (20) : $temp^{\circ}C = 0.125 \times temp_{DEC} \quad (20)$ Equations (21) and (22) show two numerical examples : $TEMP[15:0] = 0xFF38$ $temp^{\circ}C = -200 \times 0.125 = -25 \quad (21)$ $TEMP[15:0] = 0x0190$ $temp^{\circ}C = 400 \times 0.125 = 50 \quad (22)$ The temperature measurement range is [-40, 150] °C.

GPIO (0x6A)

The GPIO register is used to configure the IO0 to IO6 pins when used in GPIO mode (IOM=15).

Bits	Access	Bit Name	Default	Description
7	R/W	GPIODIR	0	Controls the direction of the digital pins (IO0 to IO6). 0: Output 1: Input
6:0	R/W	GPIO	0	If GPIODIR = 0, each bit controls the logic level of the corresponding IO pin. For example, GPIO[0] defines the level of the IO0 pin. If GPIODIR = 1 or IOM ≠ 15, each bit represents the logic level of the corresponding IO pin. For example, GPIO[0] provides the logic level of the IO0 pin. 0: Logic low 1: Logic high

READC (0x6B)

The READC register contains a counter that can be used to verify the sensor correct operation.

Bits	Access	Bit Name	Default	Description
7:0	R	READ COUNTER	0	8-bit counter that increments by 1 every time it is read. Once the maximum value, 255, is reached, it restarts from 0.


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STATUS (0x6C)

The STATUS register contains the busy flags and the general fault flags. See the Diagnostic Functions section on page 13 for more information on the diagnostic functions.

Bits	Access	Bit Name	Default	Description
7	R	MTPBUSY	0	MTP Busy flag. This flag is raised during MTP operations and it is automatically cleared when the MTP operation is completed. As long as this flag is raised, the registers from ZERO0 to PID are not accessible, neither for read nor write operations. If an operation is performed on these registers during this period, the operation is discarded and IFB flag on register IFAIL is set high. 0: MTP operation can be performed 1: MTP operation ongoing
6	-	RESERVED	-	Reserved.
5	R	INITBUSY	0	Initialization Busy flag. This flag is raised during the sensor initialization. Once the sensor is ready, the flag is automatically cleared. 0: Sensor initialization completed 1: Sensor initialization ongoing
4:3	-	RESERVED	-	Reserved.
2	R/W1C	IFAIL	0	Interface Fault flag. This flag is raised if any flag in the IFAIL register is raised. 0: No Interface Fault detected 1: Interface Fault detected This bit is cleared by writing 1. This also clears all the flags in the IFAIL register.
1	R/W1C	DFAIL	0	Digital Fault flag. This flag is raised if any flag in the DFAIL register is raised. 0: No Digital Fault detected 1: Digital Fault detected This bit is cleared by writing 1. This also clears all the flags in the DFAIL register.
0	R/W1C	AFAIL	0	Analog Fault flag. This flag is raised if any flag in the AFAIL register is raised. 0: No Analog Fault detected 1: Analog Fault detected This bit is cleared by writing 1. This also clears all the flags in the AFAIL register.


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AFAIL (0x6D)

The AFAIL register contains the analog fault flags. Each flag can be cleared by writing a 1 in the respective bit. The flag is not cleared if the fault condition is still present. See the Diagnostic Functions section on page 13 for more information on the diagnostic functions.

Bits	Access	Bit Name	Default	Description
7	-	RESERVED	-	Reserved.
6	R/W1C	OOL	0	PLL Out of Lock flag. This flag is raised if a fault condition is detected by the PLL OOL detector. 0: No fault detected 1: Fault detected
5:4	-	RESERVED	-	Reserved.
3	R/W1C	DSUPPLY	0	Digital Supply flag. This flag is raised if either an undervoltage or overvoltage is detected on the internal 1.8V supply line. 0: No fault detected 1: Fault detected
2	R/W1C	ASUPPLY	0	Analog Supply flag. This flag is raised if either an undervoltage or overvoltage is detected on the internal 3.3V supply line. 0: No fault detected 1: Fault detected
1	R/W1C	MGH	0	MagHigh flag. This flag is raised if PBI_EN = 0 and the magnetic field is above the threshold defined by the MGHT bits in the BLEVEL register. 0: Field amplitude below threshold 1: Field amplitude above threshold
0	R/W1C	MGL	0	MagLow flag. This flag is raised if PBI_EN = 0 and the magnetic field is below the threshold defined by the MGLT bits in the BLEVEL register. 0: Field amplitude above threshold 1: Field amplitude below threshold


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DFAIL(0x6E)

The DFAIL register contains the digital fault flags. Each flag can be cleared by writing a 1 in the respective bit. The flag is not cleared if the fault condition is still present. See the Diagnostic Functions section on page 13 for more information on the diagnostic functions.

Bits	Access	Bit Name	Default	Description
7:5	-	RESERVED	-	Reserved.
4	R/W1C	REGCRC	0	Register CRC flag. This flag is raised if an error is detected in the volatile memory. 0: No fault detected 1: Fault detected
3	R/W1C	MTP2	0	MTP uncorrectable error flag. This flag is raised if an uncorrectable error is detected in any MTP block during an MTP Restore operation. The volatile memory registers corresponding to blocks where an uncorrectable error is found are not updated. 0: No fault detected 1: Fault detected
2	R/W1C	MTP1	0	MTP correctable error flag. This flag is raised if a correctable error is detected in any MTP block during an MTP Restore operation. The error is corrected and the volatile memory registers corresponding to blocks where the error is found are updated. 0: No fault detected 1: Fault detected
1:0	-	RESERVED	-	Reserved.

IFAIL (0x6F)

The IFAIL register contains the interface fault flags. Each flag can be cleared by writing a 1 in the respective bit. The flag is not cleared if the fault condition is still present. See the Diagnostic Functions section on page 13 for more information on the diagnostic functions.

Bits	Access	Bit Name	Default	Description
7	R/W1C	IFB	0	Interface Busy flag. This flag is raised if a SPI/I ² C Read/Write operation is sent while an MTP operation is ongoing. 0: No fault detected 1: Fault detected
6:3	-	RESERVED	-	Reserved.
2	R/W1C	SENT	0	SENT flag. This flag is raised if a short to VDD or GND is detected on the SENT pin (see SENT Fault Detection on page 31 for more details). 0: Fault not detected 1: Fault detected
1	R/W1C	I2C	0	I ² C flag. This flag is raised if the controller if CRCEN = 1 and the controller does not send the CRC or sends an incorrect CRC is incorrect or not sent during an I ² C Write operation. 0: Fault not detected 1: Fault detected


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0	R/W1C	SPI	0	SPI flag. This flag is raised if the controller sends a wrong OpCode, CRC (if CRCEN = 1) or handshaking (if SPIHS = 1). 0: Fault not detected 1: Fault detected
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I2CSA (0x73)

The I2CSA register contains the I²C Sensor Address. See the I2C Sensor Address section on page 22 for more details.

Bits	Access	Bit Name	Default	Description
7	-	RESERVED	-	Reserved.
6:0	R	I2CSA	-	I ² C 7-bit Sensor Address defined by the logic level of the SA0, SA1 and SA2 pins on start-up. This address cannot be changed after start-up. The Sensor Address value can be obtained with Equation (23): $I2CSA = 96 + 4 \times SA2 + 2 \times SA1 + SA0 \quad (23)$ For instance, if SA2 = SA0 = 0 and SA1 = 1: $I2CSA = 96 + 4 \times 0 + 2 \times 1 + 0 = 98 \quad (24)$

ETEST (0x77)

The ETEST register is used to enable and configure the Electrical Test (ET) mode.

Bits	Access	Bit Name	Default	Description
7:4	-	RESERVED	-	Reserved.
3:1	RW	ET_SHIFT	0	Configures the reference angle generated when ET mode is enabled (ET=1). 0: 0° 1: 45° 2: 90° 3: 135° 4: 180° 5: 225° 6: 270° 7: 315° All angle values have a tolerance of δ_{ZERO} .
0	R/W	ET	0	Controls the ET mode state. 0: Disabled 1: Enabled


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MTP (0x78)

The MTP register is used to run MTP operation and check the current MTP status. See the MTP Memory Operations section on page 24 for more details.

Bits	Access	Bit Name	Default	Description
7	R	MTPDONE	1	Logic inversion of MTPBUSY flag in the STATUS register.
6	-	RESERVED	-	Reserved.
5	R/W	STORE	0	Set to 1 to store the current value of registers from ZERO0 to PID into the MTP (only performed if MTPBUSY = 0) 0: No MTP operation 1: Launch an MTP Store operation As soon as the MTP Store operation starts, the bit goes back to 0.
4	R/W	RESTORE	0	Set to 1 to copy in the registers from ZERO0 to PID the values stored in the MTP. 0: No MTP operation 1: Launch an MTP Restore operation As soon as the MTP Restore operation starts, the bit goes back to 0.
3:0	-	RESERVED	-	Reserved.

LOCKST (0x7E)

The LOCKST register contains flags related to the register lock mechanisms.

Bits	Access	Bit Name	Default	Description
7:3	-	RESERVED	-	Reserved.
2	R	MTPLOCK	1	Status of the MTP lock. When the MTP lock is enabled, the MTP register is write-locked. 0: Disabled 1: Enabled
1	-	RESERVED	-	Reserved.
0	R	WRLOCK	0	Status of the Write lock. When the Write lock is enabled, all the registers except LOCK are write-locked. 0: Disabled 1: Enabled

LOCK (0x7F)

The LOCK register is used to toggle the MTP lock and Write lock status. See the Lock Mechanisms section on page 25 for more details.

Bits	Access	Bit Name	Default	Description
7:0	R/W	LOCK	0	The MTP or Write lock status can be toggled by writing specific sequences of 3 ASCII characters in these bits.. The sequences are: 'M','T','P' (0x4D, 0x54, 0x50 in Hex) to toggle the MTP lock. 'L','C','K' (0x4C, 0x43, 0x4B in Hex) to toggle the Write lock. Any invalid character write in this register or any write in a different address cancels the sequence.



DETAILED FEATURES DESCRIPTION

Position Output Filter

The signal provided by the sensor front-end is treated by a digital filter to improve the measurement resolution and compensate the latency/lag introduced by the measurement process (see Appendix B: Definitions for the definitions of latency and lag).

Filter Window Setting

The Filter Window (FW) bits control the sensor's angle output resolution, and cutoff frequency f_{CUTOFF} .

Table 16 provides the setting options and resulting performance. When FW = 0, the digital filter is disabled.

By default (when SFEN = 1) the digital filter cancels the lag when the magnet rotation speed is constant.

Table 16: Position Output Filter Setting

FW (3:0)	τ (μs)	Resolution ⁽¹⁾ (bits)	Lag Cancellation	f_{CUTOFF} (2) (kHz)
0	0	9.8	No	100
1	2	8.0	Yes	200
2	4	8.7	Yes	125
3	8	9.4	Yes	65
4	16	10.2	Yes	29.5
5	32	10.7	Yes	13.5
6	64	11.3	Yes	6.44
7	128	11.7	Yes	3.15
8 (default)	256	12.4	Yes	1.56
9	512	12.9	Yes	0.776
10	1024	13.3	Yes	0.387
11	2048	13.8	Yes	0.193
12	4096	14.2	Yes	0.097
13	8192	14.5	Yes	0.048
14	16384	14.5	Yes	0.024
15	32768	14.5	Yes	0.012

Notes:

9) Typical value with $B_z = 48\text{mT}$ at sensing point location

10) f_{CUTOFF} defined as -3dB frequency on the response spectrum

Transfer Function

To accurately model the sensor, both the front end and the digital filter transfer functions should be considered, see Figure 50.

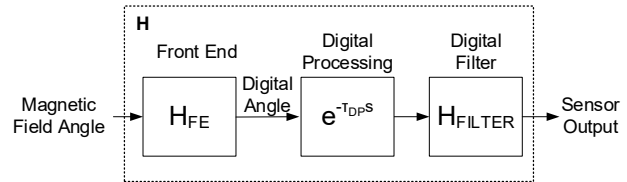


Figure 50: Transfer Function Diagram

The transfer function of the front-end signal is:

$$H_{FE} = \frac{1}{(1 + T_{FE} \times s)^4} \quad (25)$$

The transfer function of the digital filter is:

$$H_{FILTER} = \frac{1 + (2 + \delta) \times \tau \times s}{(1 + \tau \times s)^2} \quad (26)$$

The digital processing adds a propagation delay T_{DP} . The overall transfer function of the sensor is:

$$H = H_{FE} \times e^{-T_{DPS}} \times H_{FILTER} \quad (27)$$

In Equation (27) $\delta = L/T$, τ values can be found in Table 16, and T_{FE} , T_{DP} and L values are shown in General Characteristics table on page 7. If FW = 0, the filter is disabled ($H_{FILTER} = 1$).

Advanced Settings

The SFEN and LAGCEN bits give the user the possibility to change the filter behavior. If SFEN or LAGCEN is set to zero, the lag is not fully cancelled. For most applications it is recommended to use the default values SFEN = 1 and LAGCEN = 1.

Speed Output Filter

The speed output is derived from the position measurement and processed by a low-pass filter. The FWS bits in the SPEEDFILT register control resolution, time constant and cutoff frequency of the speed output, see Table 17.



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Table 17: Speed Output Filter Setting

FWS (3:0)	Resolution (bits)	Time constant τ (μ s)	f_{CUTOFF} (kHz)
0	Not recommended		
1	5.2	2	115
2	6.1	4	46.1
3	7.0	8	21.3
4	8.0	16	10.3
5	8.9	32	5.05
6	9.8	64	2.51
7	10.5	128	1.25
8 (default)	10.9	256	0.622
9	11.4	512	0.311
10	12.0	1024	0.155
11	12.7	2048	0.078
12	13.8	4096	0.039
13	14.7	8192	0.019
14	15.7	16384	0.009
15	16.0	32768	0.005

Digital Pins Configuration

IO Pins Function Selection

The function of the IO0 to IO6 pins is configured by the IOM bits in the IO0 register. Table 18 shows the configuration options.

Table 18: IO Pins Function Configuration Options

IOM	IO0	IO1	IO2	IO3	IO4	IO5	IO6
0	Pull-Down						
1	A	B	Z	U	V	W	PWM
2	A	B	Z	/A	/B	/Z	PWM
3	A	B	Z	SSCK	SSD	PBI	PWM
4	SSCK	SSD	-	U	V	W	-
5	/U	/V	/W	U	V	W	-
6	A	B	Z	-	-	-	PBI
7	-	-	-	U	V	W	-
8	MGL	MGH	-	U	V	W	-
9	-	-	SENT	-	-	-	-
10	A	B	Z	MGL	MGH	-	-
11	-	-	PWM	-	-	-	PBI
12	MGL	MGH	-	SSCK	SSD	-	-
13	TRIG	-	-	-	-	-	-
14	-	-	-	-	-	-	-
15	GPIO0	GPIO1	GPIO2	GPIO3	GPIO4	GPIO5	GPIO6

IO Pins Electrical Configuration

Pins IO0 to IO6 can be configured as push-pull or open-drain (see Figure 51) by OD012 and OD3456 bits in the IO1 register.

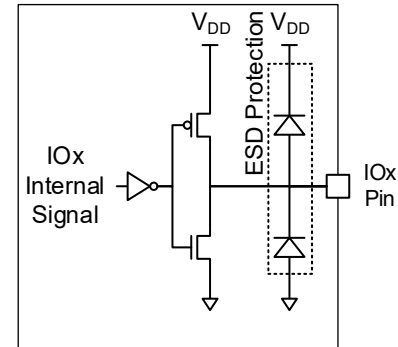


Figure 51: Internal Pin Structure for Push-Pull Configuration.

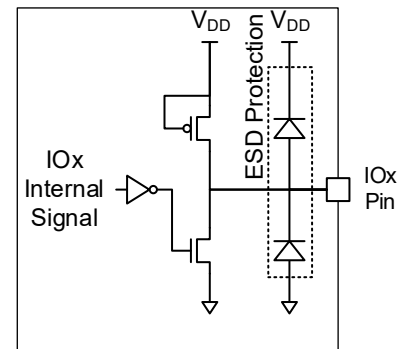


Figure 52: Internal Pin Structure for Open-Drive Configuration.

The drive strength of digital outputs is configurable by using DRV field in the IO1 register.

ABZ Incremental Encoder Output

The MA900B ABZ output emulates an incremental encoder (such as an optical encoder) providing logic pulses in quadrature (see Figure 53). Compared to signal A, signal B is shifted by a quarter of the pulse period. The number of pulses per turn, N, sent on each signal can be configured, from 1 to 2048, via the PPT bits split between the ABZ0 and ABZ1 registers.



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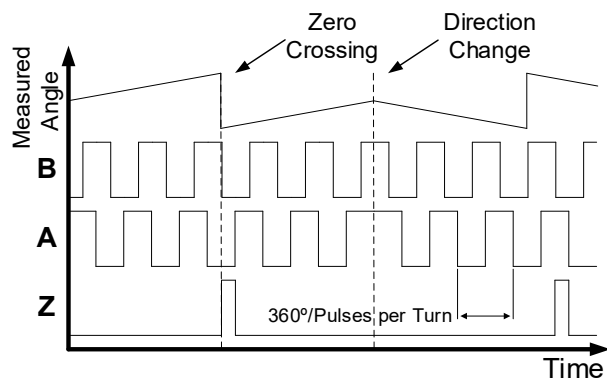


Figure 53: Timing of the ABZ Output

The Z (zero or index) signal raises only once per turn at the zero-angle position.

The position and length of the Z pulse is programmable via the ILIP bits (see Figure 54).

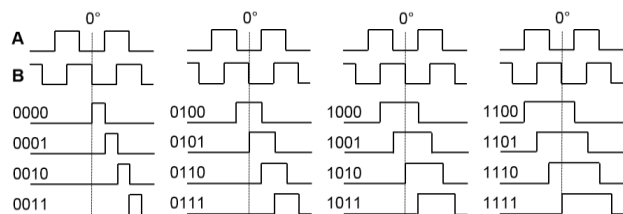


Figure 54: ILIP Effect on Index Shape

ABZ Hysteresis

The hysteresis H (Figure 55) of the A, B and Z signals can be configured via the HYST bits. To avoid spurious transitions, it is recommended to set a hysteresis larger than 12 times the sensor's RMS noise (i.e., 12σ).

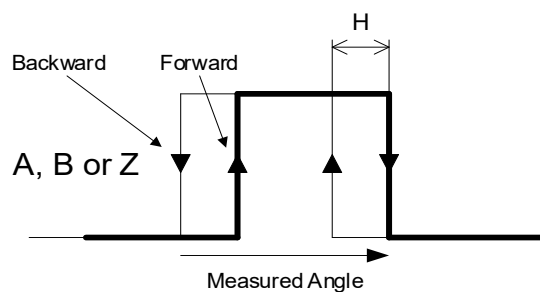


Figure 55: Hysteresis of the Incremental Output

DAZ Interface

For angular speed related applications, the ABZ interface can be converted to DAZ (D-direction / A-pulsing signal / Z-index). The B signal of ABZ is converted to D signal of DAZ. DAZ interface can be enabled by setting DAZ bit in the ABZ1 register.

In this configuration, the D signal indicates whether the angle is increasing ($D = 0$) or decreasing ($D=1$), while the A and Z signals remain unchanged compared to ABZ interface.

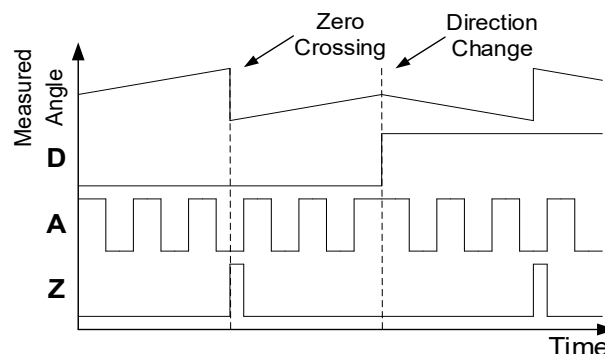


Figure 56: Timing of DAZ Output

UVW Output

The UVW output emulates the output of three Hall switches, commonly used for block commutation of 3-phase electric motors.

The three logic signals have a duty cycle, relative to the sensor measured angle, of 50% and are shifted by 120° relative to each other (see Figure 57).

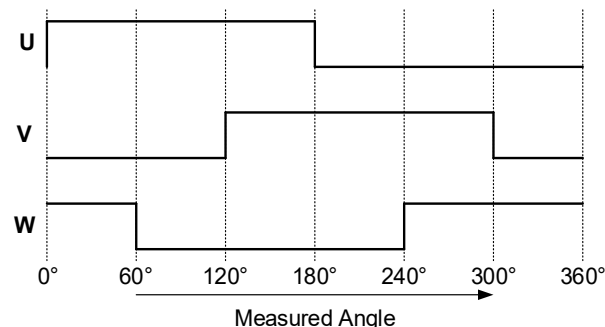


Figure 57: UVW Output

UVW Pole Pairs

The MA900B can emulate multiple pole pairs even if a target magnet with only 2 poles is used. This is achieved by dividing the digital angle into the required numbers of commutation steps per 360° revolution.

The NPP bits set the number of emulated pole pairs, and the corresponding commutation step angle for the UVW signals. Table 19 describes the pole pairs configuration options.



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Table 19: Number of UVW Pole Pairs

NPP	Emulated Pole Pairs	States per Magnetic Revolution	State Width (deg)
0	1	6	60
1	2	12	30
2	3	18	20
3	4	24	15
4	5	30	12
5	6	36	10
6	7	42	8.6
7	8	48	7.5

Figure 58 shows the UVW signals when NPP = 2 is used.

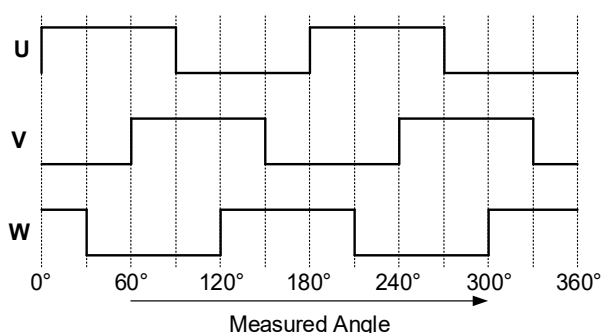


Figure 58: UVW Signals for NPP = 2

UVW Hysteresis

As for the ABZ interface, the hysteresis H (Figure 59) of the UVW signals is set by the HYST bits. It is subject to the same recommendations.

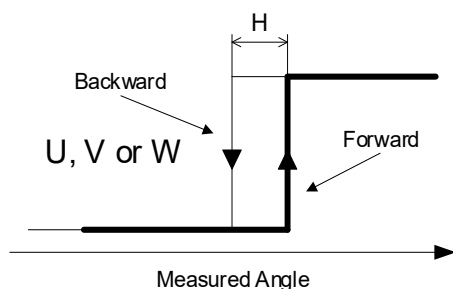


Figure 59: Hysteresis of the UVW Signal

PWM Output

The PWM output provides a logic signal whose duty cycle is a function of the measured angle.

The PWM period is divided into 4119 time units (counts). Each period is also composed by 4 segments:

- Start, the signal is held high.
- Error, the signal is held high in normal operation (Figure 60). It is held low if PWMM

= 1 and an fault condition is detected (Figure 61).

- Data, in normal operation, the signal is held high for a number of counts proportional to the measured angle, then it is driven low (Figure 60). The signal is kept low if PWMM = 1 and an fault condition is detected (Figure 61).
- End, signal is held low.

Each segment has a fixed length in counts. The length of each segment is shown in Table 20.

Table 20: PWM Segments Length in Counts

	Start	Error	Data	End	Total
Counts	12	4	4095	8	4119

The PWM interface has two configuration options:

- The PWMM bit enables the fault transmission on the PWM signal
- The PWMF bit configures the PWM signal frequency

The recommended procedure to obtain the angle value is the following:

1. Measure the PMW period t_{PWM} and the pulse width t_{HIGH} (Figure 60)
2. Calculate the length of a single count $t_{COUNT} = \frac{t_{PWM}}{4119}$
3. Calculate the pulse width in counts $C_{HIGH} = \frac{t_{HIGH}}{t_{COUNT}}$
4. The 12-bit angle is $Angle_{U12} = C_{HIGH} - 16$
5. The angle in degrees is $Angle = \frac{Angle_{U12}}{2^{12}}$



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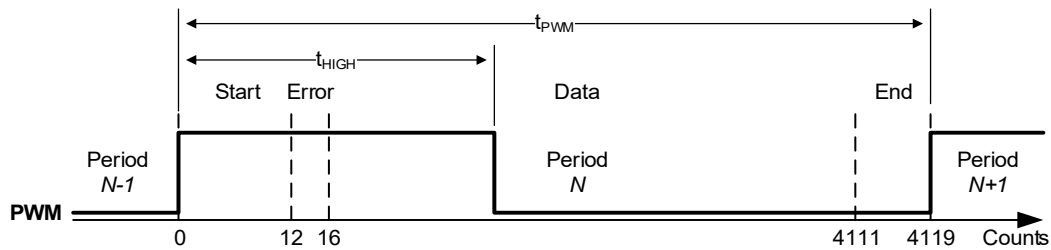


Figure 60: PWM Output, Normal Operation

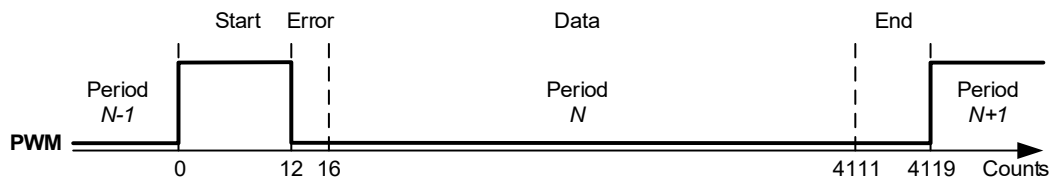


Figure 61: PWM Output when PWMM = 1 and a Fault Condition is Detected

Magnetic Field Detection Flags

MA900B provides two interfaces to monitor the field amplitude at the sensing location:

- MGH/MGL monitors if the field amplitude is inside a programmable range.
- PBI monitors the rate of change of the field amplitude, to detect events such as a button push/release.

Only one of the two interfaces can be active at a time. The user can select the active interface with the PBI_EN bit in the PBI register.

MGH/MGL

If PBI_EN = 0, the MagHigh (MGH) and MagLow (MGL) flags are enabled. They indicate that the field amplitude detected by the sensor is above (MGH) or below (MGL) a configurable threshold (see Figure 62).

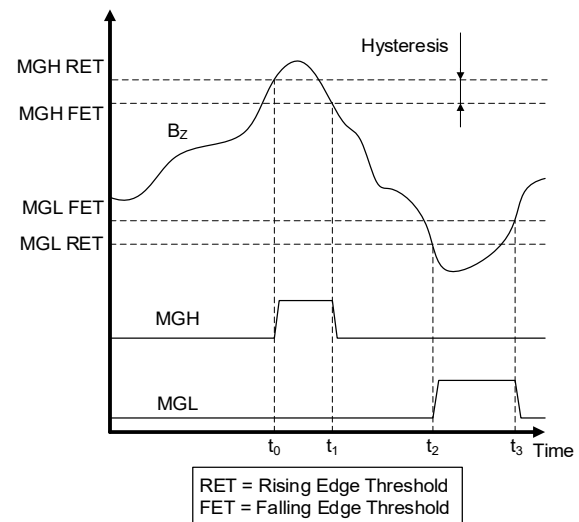


Figure 62: MGH and MGL Signals as a Function of the Field Amplitude

Each flag has its own threshold, which is configured by the MGHT and MGLT bits in the BLEVEL register (Table 21). A 3mT hysteresis is introduced between the falling and rising edges of the MGH/MGL signals.



Table 21: Magnetic Thresholds

MGxT	Magnetic field threshold [mT]		
	MGH/MGL rising edge	MGH falling edge	MGL falling edge
0000	7	4	10
0001	14	11	17
0010	21	18	24
0011	28	25	31
0100	35	32	38
0101	42	39	45
0110	49	46	52
0111	56	53	59
1000	63	60	66
1001	70	67	73
1010	77	74	80
1011	84	81	87
1100	91	88	94
1101	98	95	101
1110	105	102	108
1111	112	109	115

Every time either the MGL or MGH output is high, the nFAULT pin is driven low.

PBI

When PBI_EN = 1, the Push Button Interface (PBI) signal is enabled. The PBI signal goes high every time a large enough increase of the field amplitude (e.g. button push) is detected. On the other hand, the PBI signal goes low when a large enough decrease of the field amplitude (e.g.: button release) is detected.

To achieve this, the field amplitude signal goes through a high-pass filter, whose output is compared with configurable thresholds. The filter also inverts the signal, namely the filter output increases when the magnetic field decreases. Figure 63 shows the behavior of the PBI signal:

- At t_0 the button is pushed and the field amplitude increases
- At t_1 the high-pass filter output goes below the falling edge threshold and PBI = 1
- At t_3 the button is released, and the field amplitude decreases
- At t_4 the high-pass filter output goes above the rising edge threshold and PBI = 0

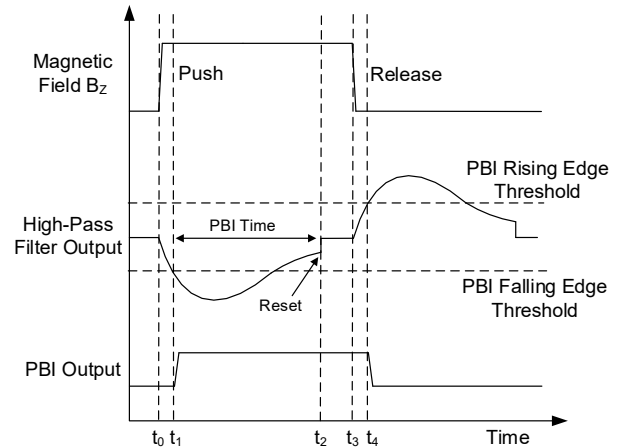


Figure 63: PBI Output

The PBI high-pass filter output is reset after a PBI edge (see t_2 in Figure 63), allowing the detection of a new event, like the button release. The time between the PBI rising edge and the high-pass filter output reset is configured by the PBI_TIME bits in the PBI register.

Like the MGH/MGL case, the magnetic threshold can be configured by the MGHT/MGLT bits in the BLEVEL register (see Table 22). MGHT configures the rising edge threshold and MGLT configures the falling edge threshold.

Table 22: PBI Thresholds

MGHT/MGLT	Rising edge [mT]	Falling edge [mT]
0000	Do not use	Do not use
0001	2.5	-2.5
0010	3	-3
0011	4	-4
0100	5	-5
0101	6	-6
0110	7	-7
0111	8	-8
1000	9	-9
1001	10	-10
1010	11	-11
1011	12	-12
1100	13	-13
1101	14	-14
1110	15	-15
1111	16	-16

**Trigger Input**

When IOM = 13, IO0 can be used as trigger input. When the trigger signal is high, the Angle, Speed, Multiturn and Temperature values are latched (not updated). This allows synchronized reading of multiple MA900B sensors.

GPIO Mode

If IOM = 15, the IO0 to IO6 pins can be used as General Purpose Input/Output pins. Pin behavior in this mode is configured by the GPIO register. The GPIODIR bit controls the direction (input or output), while the GPIO bits are used to read the input or set the output value. See the GPIO register description on page 43 for more details.



APPLICATION INFORMATION

Magnetic and Mechanical Configuration

A typical magnet is a Neodymium alloy (N35) cylinder with 6 mm diameter and 2.5 mm height, inserted into an aluminum shaft. The typical air gap between magnet and sensor is between 0.5 mm and 2.5 mm. For good measurement linearity, the sensor shall be positioned with a radial precision of typically 0.5 mm.

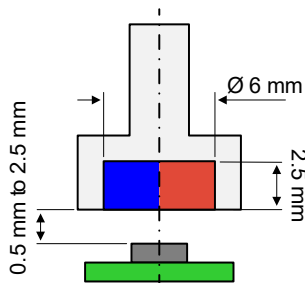


Figure 64: Typical Magnet Dimension and Placement (Cross-Section)

Magnets with either diametral or bi-axial magnetization (see Figure 65) are suitable. The two magnetization options have pros and cons: for equal magnet material, dimensions, and air gap, a bi-axial magnet generates a larger field and emits less field away, but it is less tolerant to mounting misalignments. On the other hand, a diametral magnet generates a smaller field but is more tolerant to mounting misalignments.

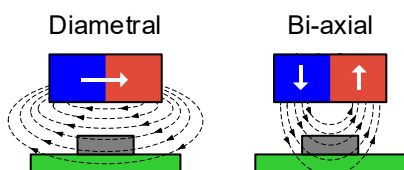


Figure 65: Diametral and Bi-axial Magnetization

Electrical Mounting and Power Supply Decoupling

For both 5V and 3.3V operation, it is recommended to place a 1 μ F decoupling capacitor, connected to the VDD pin close to the sensor with a low impedance path to the GND pin (Figure 66).

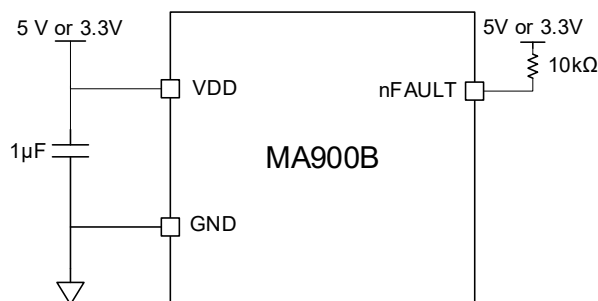


Figure 66: Connection for Supply Decoupling

Selecting the I²C Pull-Up Resistor

The I²C clock and data lines (SCL and SDA) are open drain. Pull-up resistors are required on these two lines. The resistors affect the signal edges slew rate. It is recommended to use 1.5k Ω resistors to guarantee the I²C's correct operation for all supported frequencies (see Figure 69 on page 58).

SPI Signals Routing on PCB

Typically, a direct connection between the sensor and controller SPI pins is sufficient to ensure a proper data transfer over the SPI bus. However, in specific situations (e.g.: long signal traces, external EMI presence) it is possible to improve signal integrity by a careful PCB design, especially for the SCLK line. A list of guidelines is shown below:

- Properly shield all SPI signals with a GND plane on both sides of each trace, as well as GND plane underneath the SPI signals.
- Distribute vias along these traces to connect the top and bottom GND planes.
- To avoid EMI issues, route the SCLK signal away from the other SPI signals and noise sources. The distance should be at least three times the SCLK trace width.
- Insert an RC low-pass filter on SCLK (see Figure 67 on page 57) close to the sensor. It is recommended to use a 200 Ω serial resistor with a 10pF capacitor to GND (80MHz cutoff frequency).
- Use a star topology for the GND connection and keep it as direct and short as possible to avoid ground loops.



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- Insert RC low-pass filters on the CIPO and COPI signals (see Figure 67). The RC filter on COPI must be located close to the controller, while the filter on CIPO must be located close to the sensor. It is recommended to use a 200Ω resistor with a 10pF capacitor.
- Avoid a significant trace length mismatch between the SPI signals, especially between the CIPO, COPI, and SCLK signals. These signals should have similar propagation delays.
- If possible, avoid placing vias on the SCLK signal.

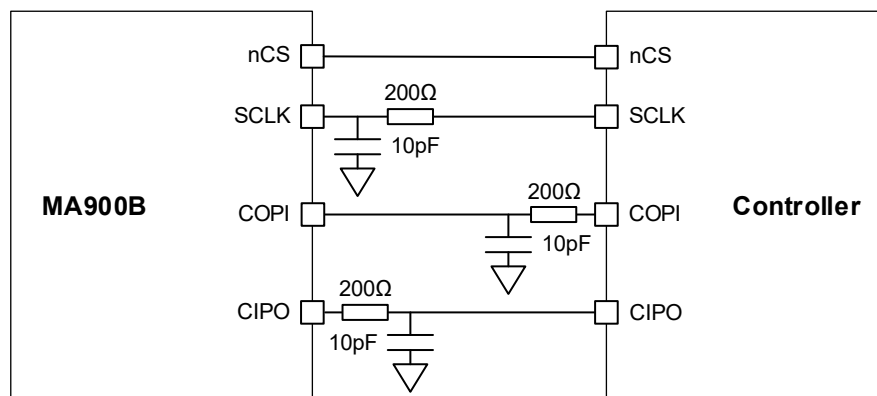


Figure 67: Examples of RC Low-Pass Filters on the SPI Signals



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TYPICAL APPLICATION CIRCUIT

SPI

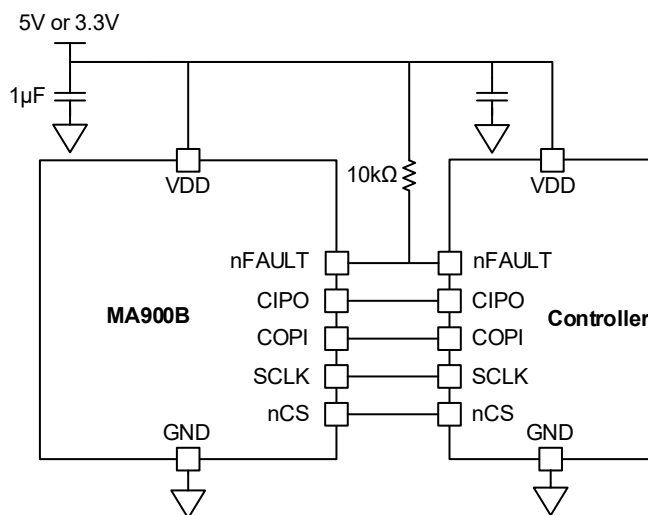


Figure 68: Typical Application Circuit - SPI

I²C

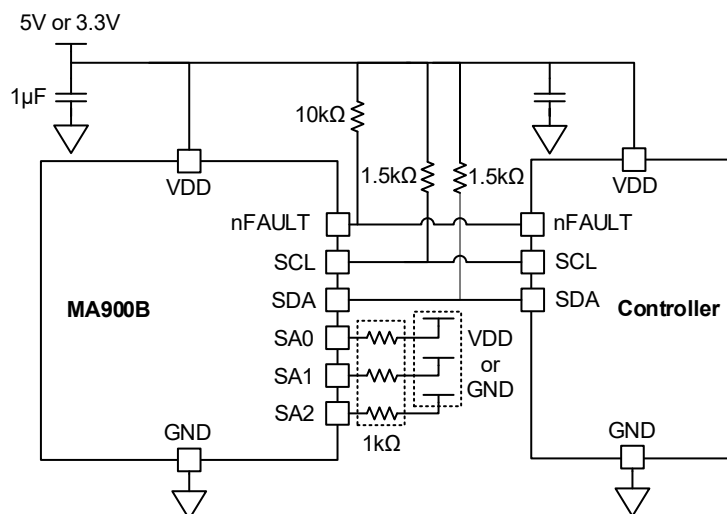
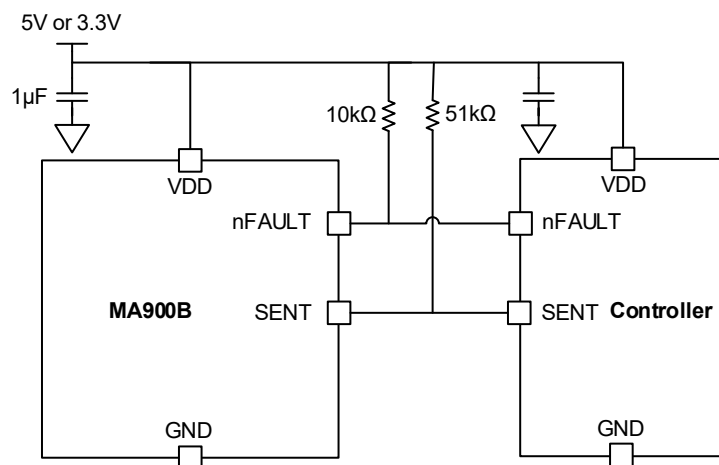
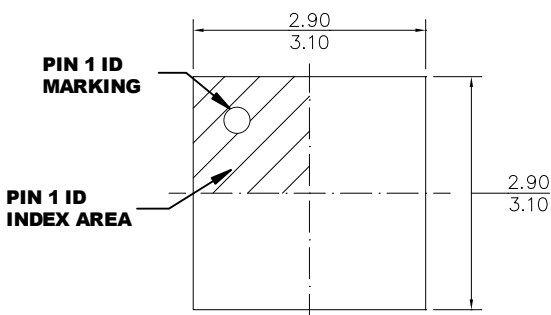
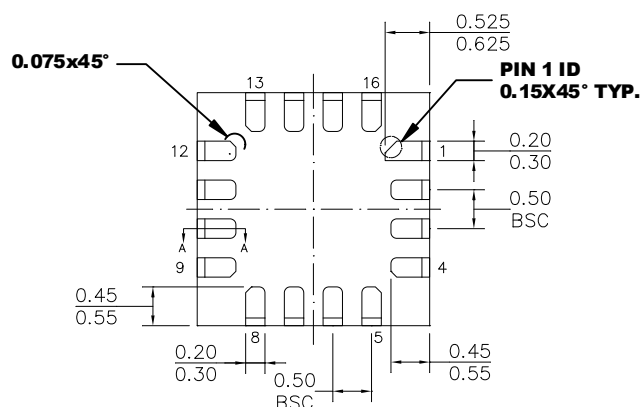
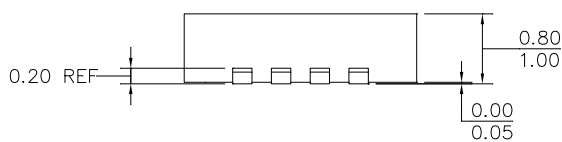
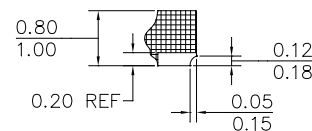
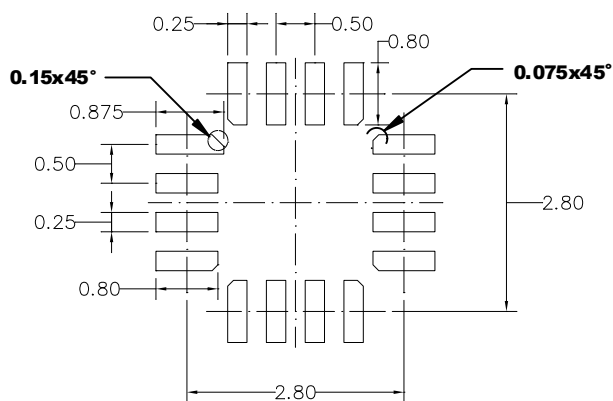


Figure 69: Typical Application Circuit - I²C

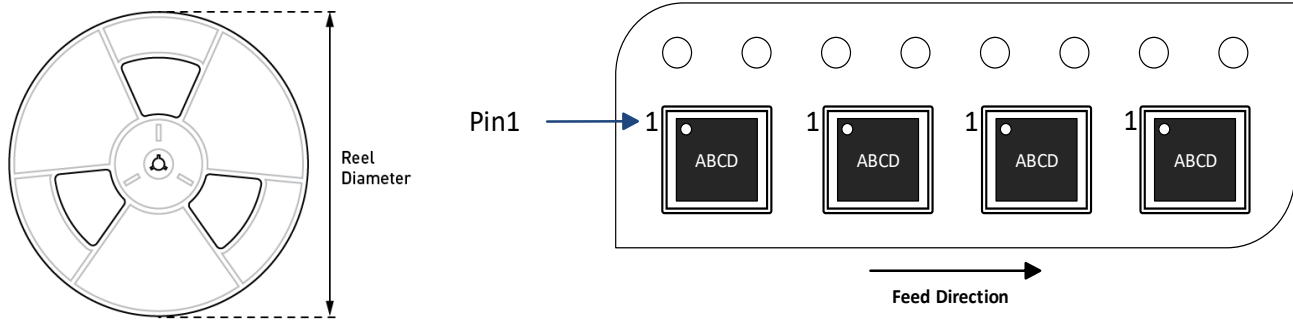

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SENT

Figure 70: Typical Application Circuit - SENT



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PRELIMINARY SPECIFICATION SUBJECT TO CHANGE**PACKAGE INFORMATION****QFN-16 (3mmx3mm)****TOP VIEW****BOTTOM VIEW****SIDE VIEW****SECTION A-A****RECOMMENDED LAND PATTERN****NOTE:**

- 1) THE LEAD SIDE IS WETTABLE.**
- 2) ALL DIMENSIONS ARE IN MILLIMETERS.**
- 3) LEAD COPLANARITY SHALL BE 0.08 MILLIMETERS MAX.**
- 4) JEDEC REFERENCE IS MO-220.**
- 5) DRAWING IS NOT TO SCALE.**


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CARRIER INFORMATION


Part Number	Package Description	Quantity/ Reel	Quantity/ Tube	Quantity/ Tray	Reel Diameter	Carrier Tape Width	Carrier Tape Pitch
MA900BGQE-C-0000-Z	QFN-16 (3mm x 3mm)	5000	N/A	N/A	13 in.	12 mm	8 mm
MA900BGQE-S-0000-Z							
MA900BGQE-C-xxxx-Z							
MA900BGQE-S-xxxx-Z							



APPENDIX A: CRC CALCULATION

SPI CRC4

The CRC polynomial used for SPI is $x^4+x^3+x^2+1$, with an initial value of 10.

The C code below shows how to calculate the 16-bit value to send on COPI by appending the 4-bit CRC to the 12-bit data using a pre-computed CRC lookup table.

```
/* pre-computed CRC table for x^4+x^3+x^2+1 */
const uint8_t crc4_table[16] = {0,13,7,10,14, 3,9,4,1,12,6,11,15,2,8,5};
uint16_t append_crc4( uint16_t data ) {           // data is only 12-bit
    uint8_t crc4;                                // 4-bit CRC (0-15)
    // Compute CRC4
    crc4 = 10;                                   // Initial Value for CRC4 (Seed)
    crc4 = (data>>8 & 0xF) ^ crc4_table[crc4];   // 4 MSB first
    crc4 = (data>>4 & 0xF) ^ crc4_table[crc4];   //
    crc4 = (data>>0 & 0xF) ^ crc4_table[crc4];   // 4 LSB

    return (data<<4) | crc4;                     //Concatenate 12-bit data with CRC4
}
```

Figure 71 shows an example of an SPI Read of Register 17. The CIPO frames are shown both for regular and inverted CRC cases.

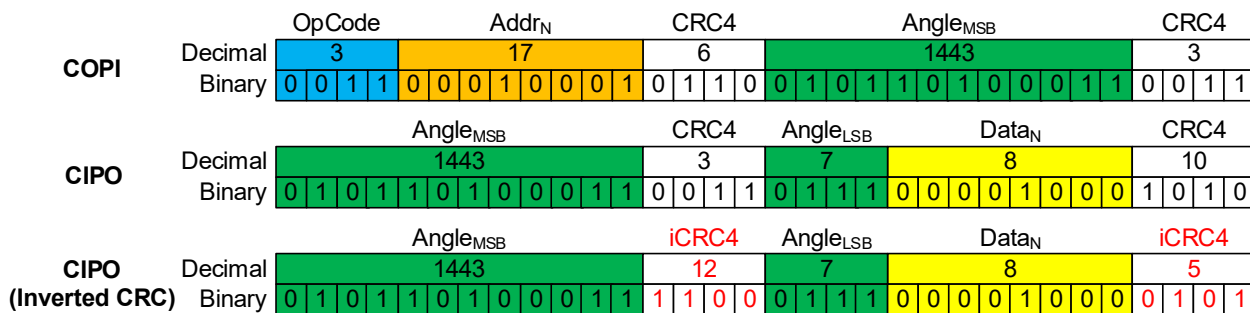


Figure 71: SPI Read Example with CRC values

I²C CRC8

The CRC polynomial used for I²C is $x^8+x^2+x^1+x^0$, with an initial value of 0.

The C code below shows how to calculate the 8-bit CRC using a pre-computed CRC lookup table.

```
/* pre-computed CRC table for x^8+x^2+x^1+x^0 */
const uint8_t crc8_table [256] = {
    0x00,0x07,0x0e,0x09,0x1c,0x1b,0x12,0x15,0x38,0x3f,0x36,0x31,0x24,0x23,0x2a,0x2d,
    0x70,0x77,0x7e,0x79,0x6c,0x6b,0x62,0x65,0x48,0x4f,0x46,0x41,0x54,0x53,0x5a,0x5d,
    0xe0,0xe7,0xee,0xe9,0xfc,0xfb,0xf2,0xf5,0xd8,0xdf,0xd6,0xd1,0xc4,0xc3,0xca,0xcd,
    0x90,0x97,0x9e,0x99,0x8c,0x8b,0x82,0x85,0xa8,0xaf,0xa6,0xa1,0xb4,0xb3,0xba,0xbd,
    0xc7,0xc0,0xc9,0xce,0xdb,0xdc,0xd5,0xd2,0xff,0xf8,0xf1,0xf6,0xe3,0xe4,0xed,0xea,
    0xb7,0xb0,0xb9,0xbe,0xab,0xac,0xa5,0xa2,0x8f,0x88,0x81,0x86,0x93,0x94,0x9d,0x9a,
    0x27,0x20,0x29,0x2e,0x3b,0x3c,0x35,0x32,0x1f,0x18,0x11,0x16,0x03,0x04,0x0d,0x0a,
    0x57,0x50,0x59,0x5e,0x4b,0x4c,0x45,0x42,0x6f,0x68,0x61,0x66,0x73,0x74,0x7d,0x7a,
    0x89,0x8e,0x87,0x80,0x95,0x92,0x9b,0x9c,0xb1,0xb6,0xbf,0xb8,0xad,0xaa,0xa3,0xa4,
    0xf9,0xfe,0xf7,0xf0,0xe5,0xe2,0xeb,0xec,0xc1,0xc6,0xcf,0xc8,0xdd,0xda,0xd3,0xd4,
    0x69,0x6e,0x67,0x60,0x75,0x72,0x7b,0x7c,0x51,0x56,0x5f,0x58,0x4d,0x4a,0x43,0x44,

```



```
0x19,0x1e,0x17,0x10,0x05,0x02,0x0b,0x0c,0x21,0x26,0x2f,0x28,0x3d,0x3a,0x33,0x34,
0x4e,0x49,0x40,0x47,0x52,0x55,0x5c,0x5b,0x76,0x71,0x78,0x7f,0x6a,0x6d,0x64,0x63,
0x3e,0x39,0x30,0x37,0x22,0x25,0x2c,0x2b,0x06,0x01,0x08,0x0f,0x1a,0x1d,0x14,0x13,
0xae,0xa9,0xa0,0xa7,0xb2,0xb5,0xbc,0xbb,0x96,0x91,0x98,0x9f,0x8a,0x8d,0x84,0x83,
0xde,0xd9,0xd0,0xd7,0xc2,0xc5,0xcc,0xcb,0xe6,0xe1,0xe8,0xef,0xfa,0xfd,0xf4,0xf3,
};
```

```
uint8_t crc8(uint8_t crc8_curr, uint8_t data) { // inputs are current CRC value and data
    crc8 = crc8_table[crc8_curr ^ data]; //table index is XOR between current CRC and data
    return crc8;
```

I²C Read

The following code shows how the controller must calculate the CRC for a single I²C Read:

```
uint8_t sensor_address = 0x60; // sensor i2c address
uint8_t reg_address = 0x00; // address of the register where the write is
// performed
uint8_t reg_value = 0x05; // value to write
uint8_t crc = 0;
crc = crc8(crc, sensor_address << 1);
crc = crc8(crc, reg_address);
crc = crc8(crc, sensor_address << 1 + 1);
crc = crc8(crc, reg_value);
```

If read burst is used, CRC is sent after each register value. Assuming the registers' values are stored in an array, for each iteration *i*, the CRC is calculated as follows:

```
uint8_t crc = 0;
crc = crc8(crc, sensor_address << 1);
crc = crc8(crc, reg_address + i);
crc = crc8(crc, sensor_address << 1 + 1);
crc = crc8(crc, write_value[i]);
```

If quick read is used, CRC is also sent after each register value. Assuming the registers' values are stored in an array, for each iteration *i*, the CRC is calculated as follows:

```
uint8_t crc = 0;
crc = crc8(crc, sensor_address << 1 + 1);
crc = crc8(crc, 0x60 + i); // 0x60 is the first register accessed by quick read
crc = crc8(crc, reg_value[i]);
```

I²C Write

The following code shows how the controller must calculate the CRC for a single I²C Write:

```
uint8_t sensor_address = 0x60; // sensor i2c address
uint8_t reg_address = 0x00; // address of the register where the write is
// performed
uint8_t write_value = 0x05; // value to write
uint8_t crc = 0;
crc = crc8(crc, sensor_address << 1);
crc = crc8(crc, reg_address);
crc = crc8(crc, write_value);
```

If write burst is used, CRC must be sent after each write value. Assuming the write values are stored in an array, for each iteration *i*, the CRC is calculated as follows:

```
uint8_t crc = 0;
crc = crc8(crc, sensor_address << 1);
crc = crc8(crc, reg_address + i);
```



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```
crc = crc8(crc, write_value[i]);
```

Numerical Example

Figure 72 shows an example of an I²C Single Read of Register 115. The data returned by the sensor is 96 (the SA0, SA1 and SA2 pins are connected to ground at power-up).

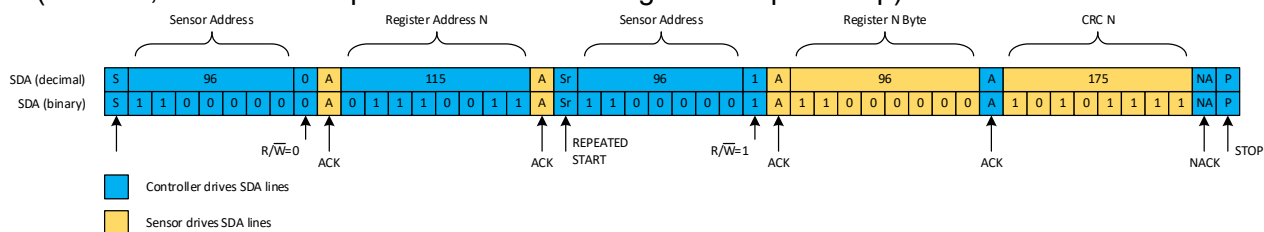


Figure 72: I2C Read of Register I2CSA for SA2,1,0 = 0



APPENDIX B: DEFINITIONS

Resolution (3σ noise level)	Smallest angle increment distinguishable from the noise. The resolution is measured by computing three times σ (the standard deviation in degrees) taken over 1,000 data points at a constant position. The resolution in bits is obtained with $\log_2(360 / 6\sigma)$.
Refresh Rate	Rate at which new data points are stored in the output buffer.
ABZ Update Rate	Rate at which a new ABZ state is computed. The inverse of this rate is the minimum time between two ABZ edges.
ABZ Jitter	The jitter characterizes how far a particular ABZ edge can occur at an angular position different from the ideal position (see Figure B1). It does not include the systematic measurement error.

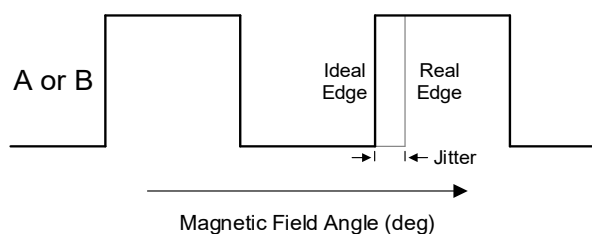


Figure B1: ABZ Jitter

The measurable jitter is composed of a systematic jitter (always the same deviation at a given angle) and a random jitter.

At very low speeds, the random jitter matches the angle measurement noise. As the speed gets higher, the random jitter becomes smaller than the angle measurement noise. This is because, at high speed, the low frequency part of the noise spectrum does not contribute to the jitter.

Latency	Time elapsed between the instant when the data is ready to be read and the instant at which the shaft passes that position.
Lag	The lag (in degrees) is $Lag = Latency \times v$, where v is the angular velocity (in deg/s).
Start-Up Time	Time until the sensor delivers valid data, starting at power-up.



MA900B – ANGLE SENSOR WITH STRAY-FIELD IMMUNITY

PRELIMINARY SPECIFICATION SUBJECT TO CHANGE

Integral Nonlinearity (INL)

Maximum deviation between the average sensor output (at a fixed position) and the true mechanical angle (see Figure B).

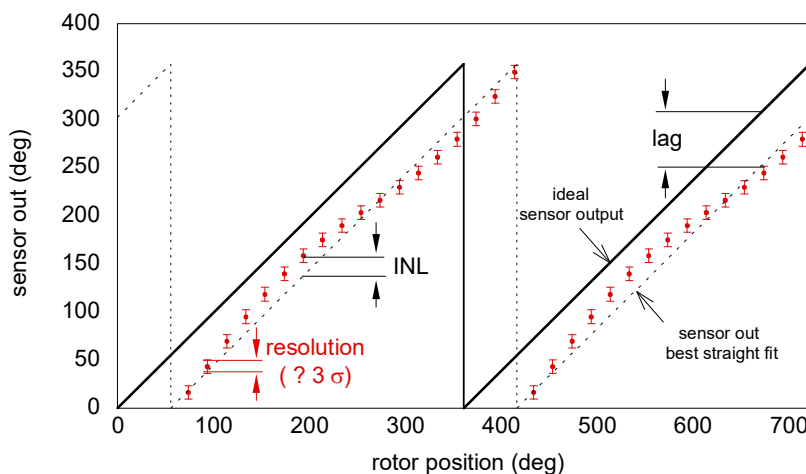


Figure B2: Resolution, INL, Lag

INL can be obtained from the error curve $err(\alpha) = out(\alpha) - \alpha$, where $out(\alpha)$ is the average sensor output calculated over many samples (e.g. 1000), and α is the mechanical angle indicated by a high-precision ($<0.001^\circ$) encoder. INL is then calculated with Equation (B1):

$$INL = \frac{\max (err(\alpha)) - \min (err(\alpha))}{2} \quad (B1)$$



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INL Harmonics

Since the error is a periodical function of period 2π , it can be expressed as a Fourier Series. The formula is shown in Equation B2:

- $err(\alpha)$ is the error curve.
- α is the mechanical angle in radians.
- Each n^{th} element of the summation is referred to as the n^{th} harmonic component of the error.
- H_0 is the offset between the sensor output and the mechanical angle, namely the mean value of $err(\alpha)$ (see Equation B3).
- H_n is the amplitude of the n^{th} harmonic component (see Equation B4).
- φ_n is the phase in radians of the n^{th} harmonic component (see Equation B5).

$$err(\alpha) = H_0 + \sum_{n=1}^N H_n \cos(n\alpha - \varphi_n) \quad (B2)$$

$$H_0 = \frac{1}{2\pi} \int_0^{2\pi} err(\alpha) d\alpha \quad (B3)$$

$$H_n = \sqrt{A_n^2 + B_n^2} \quad (B4)$$

$$\varphi_n = \tan^{-1} \left(\frac{B_n}{A_n} \right) \quad (B5)$$

$$A_n = \frac{1}{\pi} \int_0^{2\pi} err(\alpha) \cos(n\alpha) d\alpha \quad (B6)$$

$$B_n = \frac{1}{\pi} \int_0^{2\pi} err(\alpha) \sin(n\alpha) d\alpha \quad (B7)$$

Angle Drift

Worst rate of change, over all angles, when one parameter is changed (e.g. temperature, VDD) and all the others, including the B field angle, are kept constant.

Angle Variation

Worst total change, over all angles, when one parameter is changed (e.g. temperature, VDD) and all the others, including the B field angle, are kept constant.

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